

Related Party Trade and Gravity: Revisiting the Distance Effects

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Abstract

This paper investigates how internal firm distance costs shape the geography of foreign direct investment (FDI) and the export behavior of foreign multinational affiliates. Using shipment-level export data from India, I document how related-party trade patterns have evolved over the past decade. The data show that foreign affiliates engage in export behavior that is more complex than what is typically captured in the existing FDI literature and that these patterns have shifted notably in recent years. Motivated by these facts, I derive a firm-level gravity equation with heterogeneous distance elasticities for related-party and arm's-length exports and estimate it using Poisson pseudo-maximum likelihood. The results indicate that related-party exports are less sensitive to distance than arm's-length exports, consistent with the distance premium for within-firm trade documented in domestic settings. Importantly, I also find that this internal distance premium has eroded in recent years, helping to explain the recent rise in inter-firm trade among the foreign affiliates of multinational firms.

Keywords: Multinational enterprise (MNE), foreign MNE affiliate, related-party trade, firm-level gravity, distance elasticity

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1 Introduction

Foreign direct investment (FDI) is a central driver of globalization, providing a key channel through which multinational enterprises (MNEs) expand production across borders.¹ FDI and multinational activity shape labor markets, productivity, and innovation,² underscoring the importance of understanding how MNEs organize their activities across space. The geography of FDI is commonly analyzed through two lenses: horizontal FDI, which substitutes for exports by replicating production abroad, and vertical FDI, which fragments production across countries and complements trade. Empirical and theoretical work, including [Irrarrazabal et al. \(2013\)](#) and [Bartelme and Ziv \(2024\)](#), suggests that within-firm supply chains matter for the geography of multinational activity and that multinational firms appear to face within-firm distance costs. However, no estimates of within-firm overseas trade costs currently exist to substantiate these hypotheses. Moreover, despite extensive documentation of MNEs' FDI decisions, the structure and nature of MNE trade within this global production geography remain opaque.

This paper examines the evolving role of internal firm distance costs in shaping the geography of FDI and the export behavior of foreign MNE affiliates. Using shipment-level export data from India, I construct a unique dataset that identifies whether each transaction is related-party (within-firm) or arm's-length (across-firm) over the past decade.³ This study presents a new method for identifying within-firm trade, making it possible to better observe affiliate-to-affiliate transactions. Most firm-level datasets widely used in the literature on FDI and MNEs have limited coverage.⁴ Many of these confidential administrative datasets identify related-party trade using ownership information, including the presence and share of foreign ownership. In this study, I use Bill of Lading data, which are publicly accessible, to match shipping and receiving firms and classify transactions as either within-firm or across-firm. Using information on each exporting firm's ultimate parent and the number of countries in which that parent operates, I first determine whether the ultimate parent qualifies as an MNE and then identify the corresponding foreign affiliates by matching exporter and importer names. Each shipment is classified as either an arm's-length or a related-party transaction, including both affiliate-to-parent and affiliate-to-affiliate shipments. Crucially, my matching procedure for identifying sister

¹According to the IMF, the global stock of inward FDI exceeded USD 41 trillion in 2023—roughly two-fifths of world GDP. MNEs account for about one-third of global output ([Delpeuch et al., 2025](#)).

²They influence employment ([Lipsey and Sjöholm, 2004](#)), wages ([Setzler and Tintelnot, 2021](#); [Alfaro-Urena et al., 2021](#)), mobility ([Balsvik, 2011](#)), productivity ([Smarzynska Javorcik, 2004](#); [Alfaro and Chen, 2018](#)), product quality ([Ge et al., 2015](#)), and knowledge diffusion ([Keller, 2010](#)); see also [Moran et al. \(2005\)](#); [Moran \(2011\)](#).

³Throughout the paper, I use the terms related-party and within-firm interchangeably, as well as arm's-length, between-firm, and across-firm. The term 'intrafirm' specifically refers to trade between a parent firm and its affiliate (or vice versa), unless otherwise indicated by the context.

⁴The majority of existing studies rely on data from the United States, while others use datasets from selected European countries (e.g., Belgium ([Conconi et al., 2025](#)), France ([Davies et al., 2018](#))), Latin America (e.g., Costa Rica ([Alfaro-Ureña et al., 2022](#))), and Asia (e.g., China ([Wang, 2021](#))).

establishments in both home and foreign countries does not rely on ownership relations and is therefore better suited to documenting affiliate-to-affiliate transactions.⁵

Using the data to examine the shipment-level trade activity of foreign MNE affiliates in detail, I document four stylized facts about their export behavior. First, as documented in [Ramondo et al. \(2016\)](#), I confirm that foreign MNEs sell through both channels—related-party (within-firm) and arm’s-length (across-firm)—to foreign markets via their affiliates in India. Second, I show that they mainly sell intermediate inputs and typically export distinct sets of products across the two modes. These facts imply that foreign affiliates of MNEs participate in both within-firm and across-firm supply chains. Third, the probability of engaging in related-party exports is positively correlated with distance. Fourth, related-party exports were disproportionately directed toward distant markets through 2020, but this pattern has weakened in the past few years. Taken together, these facts suggest that the distance elasticity of exports may differ between internal trade within MNE networks and arm’s-length trade and that this gap may have narrowed in recent years.

Guided by these facts, I construct a partial-equilibrium framework grounded in the Ricardian model of [Eaton and Kortum \(2002\)](#). A large set of suppliers produce and sell a continuum of goods and can serve foreign markets either through related-party or arm’s-length exports, with trade costs that differ across these two modes. The equilibrium conditions yield estimable firm-level trade equations that capture both within- and across-firm distance elasticities. I estimate the model using a Poisson pseudo-maximum likelihood estimator. I implement the estimation using Bill of Lading data from India and Mexico to align the empirical implementation with the theoretical model, which allows for the inclusion of destination-country fixed effects. The results indicate that related-party exports are less sensitive to distance than arm’s-length exports, but that this difference has indeed diminished over time. Within the context of the model, this erosion of distance elasticities helps explain the recent expansion of cross-firm trade among MNEs: as the relative advantage of within-firm trade at a distance declines, MNE affiliates increasingly engage in arm’s-length transactions abroad.

The contributions of this study are threefold. The first contribution is to the literature on identifying within-firm trade among MNEs. A key challenge in the empirical analysis of multinational firm activity has been the difficulty of observing both within- and across-firm trade. Earlier seminal work concluded that within-firm trade is relatively limited or sparse among U.S. MNEs ([Ramondo et al., 2016](#)). However, [Alfaro et al. \(2025\)](#) raises concerns about potential undercounting in U.S. MNE data, proposes a methodological remedy, and finds that within-firm trade between parents and affiliates is, in fact, prevalent. [Alfaro et al. \(2025\)](#) also notes that other studies—particularly those using U.S. Customs

⁵ [Alfaro et al. \(2025\)](#) point out that the Longitudinal Firm Trade Transaction Database (LFTTD), the most widely used source on MNE activity in the United States, flags related-party trade based on ownership shares. Further details are provided in Footnote 6.

data—document parent–affiliate trade, but a persistently unaccounted-for aspect of within-firm trade is affiliate-to-affiliate transactions.⁶ Although recent papers, such as [Antràs et al. \(2024\)](#), combine multiple data sources, important limitations remain. In this study, I use publicly accessible transaction-level data and match exporting and importing firm names to identify related-party trade. By doing so, I am able to observe affiliate-to-affiliate transactions and incorporate them into the broader measure of related-party trade. A related study, [Hong \(2023\)](#), also begins to address this gap by developing a new proxy and methodology to identify related-party trade more broadly using web-scraped financial statements of firms in South Korea.

This study’s second contribution is to the literature documenting the trade behavior of foreign affiliates of MNEs. [Cadestin et al. \(2018\)](#) document that foreign affiliates are often highly export-oriented, yet affiliate trade beyond transactions with the parent firm remains underexplored. A growing body of work has examined affiliate-level export patterns using firm-level data. For example, [Garetto et al. \(2024\)](#) show that foreign affiliates of U.S. MNEs typically begin with host-market sales before expanding into exports and that their market entry patterns are largely unaffected by sibling affiliates in other countries.⁷ Another recent paper by [Conconi et al. \(2025\)](#) shows that firms acquired by MNEs boost affiliates’ participation in export markets primarily by alleviating market-specific entry frictions. In addition, [Antràs et al. \(2024\)](#) and [Wang \(2021\)](#) highlight the correlation between a firm’s location and export destination, showing that affiliates tend to trade more with countries geographically closer to the location of the foreign affiliate (plant) or headquarters. Building on this line of research, my study focuses on the transaction channel—within or across firms—rather than on the timing or geographic scope of exports. This study is closely related to analyses of how trade is organized within multinational firms. [Ramondo et al. \(2016\)](#) show that foreign affiliates of U.S. MNEs sell largely outside the firm, a result also documented in [Antràs et al. \(2024\)](#). Similarly, [Davies et al. \(2018\)](#) find limited within-firm trade among MNEs in France. I confirm these results using data on the universe of MNE affiliate transactions in India and capture affiliate-to-affiliate exports, including those with no direct ownership relations between the two trading parties. I

⁶The most widely used source of data on MNE activity in the United States is the Longitudinal Firm Trade Transaction Database (LFTTD), which is based on U.S. Customs data. For U.S. exports in this dataset, a related-party transaction is defined in the Foreign Trade Regulations, 15 C.F.R. §30.1, as “a transaction involving trade between a U.S. principal party in interest and an ultimate consignee where either party owns directly or indirectly 10 % or more of the other party.”. Another dataset, the Related Party Trade data published by the U.S. Census Bureau, uses the same definition ([Ruhl, 2015](#)). This definition narrowly captures related-party trade and essentially implies a parent–affiliate relationship. An exception to this may be studies using Bureau of Economic Analysis (BEA) survey data. The BEA provides information on affiliates’ sales by transactor and destination; however, the BEA data consolidate information across all foreign affiliates, and shipment-level information on affiliate-to-affiliate trade is not available. As a result, sector- or product-level details are lost. Moreover, as discussed by [Alfaro et al. \(2025\)](#), the BEA data may undercount affiliate-to-affiliate transactions.

⁷They use a long panel of U.S. MNE data from the BEA spanning 1987–2011, focusing on affiliate activity in the top ten U.S. FDI destinations.

further extend these results by leveraging the product-level detail in my data, showing that affiliates export distinct sets of products within and across firms. Moreover, I show how selling outside the firm has become increasingly dominant over time.

Finally, I contribute to a growing literature on distance frictions within firms and their implications for the geography of within-firm trade. Several studies examine the link between proximity and within-firm trade in domestic settings. [Garg et al. \(2023\)](#) document a strong preference for sourcing from within-firm suppliers in India over distance, driven by investment specificity. [Atalay et al. \(2019\)](#) show that U.S. firms are more likely to conduct internal shipments at any given distance, suggesting a distance premium associated with common ownership. Consistent with this, [Bartelme and Ziv \(2024\)](#) find that larger U.S. firms operate production networks that are less geographically concentrated because they benefit from lower distance costs, while average firms remain more spatially clustered. However, whether these domestic patterns extend to cross-border within-firm trade by MNEs remains an open question. Existing evidence on distance in multinational production largely stems from seminal models predicting strong distance effects on foreign affiliates' sales in the host country due to input trade between the parent and affiliates⁸ ([Irrazabal et al., 2013](#); [Antràs and Yeaple, 2014](#); [Bombarda and Marcassa, 2020](#); [Alviarez and Saad, 2023](#)). Yet we know much less about within-firm trade beyond this margin—in particular, within-firm shipments among foreign affiliates across host countries—and about how these patterns compare with arm's-length trade across the same geography. I move beyond this traditional focus in two ways. First, using shipment-level data, I show that distance systematically shapes MNEs' within- versus across-firm sourcing behavior differently. Second, I estimate and compare the distance elasticities of within- and across-firm trade using data from two countries, India and Mexico, providing direct evidence of differential distance costs within multinational production networks. Together, these findings reveal a distance premium in within-firm overseas trade, consistent with the pattern documented for domestic within-firm shipments, while also showing notable changes in recent years.

The remainder of the paper is organized as follows. Section 2 describes the data and explains how I identify foreign MNE affiliates and related-party trade. Section 3 presents stylized facts on the export behavior of foreign MNE affiliates. Section 4 outlines the theoretical framework and derives the estimable firm-level trade equation. Section 5 describes the empirical implementation. Section 6 presents the main results. Section 7 discusses the findings and considers potential mechanisms behind the evolving difference in distance elasticities. Section 8 concludes.

⁸That is, affiliates in the host country source inputs from parent firms located in the country where the parent is based

2 Data

In this section, I introduce the Bill of Lading data, outline the data processing procedures, and describe the additional datasets used in this study.

2.1 Bill of Lading and S&P Firm Data

Bills of Lading (BoL) are essential legal instruments in global shipping and logistics that provide detailed information on physical trade transactions between exporting and importing firms. BoL data have been widely used to study trade shocks and recovery (Flaen et al., 2023), trade routes and indirect shipping costs (Ganapati et al., 2024), transshipment (Do et al., 2025), and the links between financial globalization and international trade (Bruno and Shin, 2023), among other topics.

This study uses BoL data for India obtained from the S&P Global Panjiva Supply Chain Intelligence database (hereafter S&P Panjiva), complemented with data on firms' corporate structures.⁹ The BoL records provide detailed information on exporting and importing firms—including firm names and unique identifiers—traded products (identified by Harmonized System codes), shipment destinations, and the value of goods, covering the period from 2016 to 2024.

I use the Indian and Mexican data for two reasons. First, BoL records in both countries are reported at the item level rather than solely at the shipment level.¹⁰ This item-level structure enables precise product identification and allows the construction of firm-product pairs as the unit of analysis. Second, both India and Mexico are economically important locations for multinational activity.¹¹

For the stylized facts in Section 3, I focus on India because the processed Indian sample remains sufficiently rich and representative for documenting broad regularities. The Mexican sample, by contrast, loses many observations during cleaning and is only partially available in the final sample, which makes it less suitable for documenting stylized patterns. I nevertheless retain the Mexican observations in the regression analysis because they help strengthen identification.

In the following subsections, I describe the data processing procedures and summarize the dataset. To keep the exposition clear, I focus on the construction of the Indian data.

⁹Sources: S&P Capital IQ Base and S&P Ultimate Parent.

¹⁰A single shipment may contain multiple items or product lines; thus, using only shipment-level data can obscure product-level variation.

¹¹In 2023, inward FDI inflows reached \$28.2 billion in India and \$36.1 billion in Mexico. Their inward FDI stocks were also large, at \$536.9 billion and \$778.4 billion, respectively. Moreover, both countries recorded an increase of about \$130 billion in inward direct investment positions in 2023, or roughly 20 %, underscoring their importance as host economies for multinational firms (UNCTAD, 2024; Leino and Gavrilovic, 2025).

2.1.1 Identifying MNEs and Foreign MNE Affiliates

First, I link firms in the BoL data to S&P's firm-level data on corporate structure. These data provide information on the ultimate parent of firms in the BoL records.¹² I count the number of countries in which each ultimate parent has affiliated entities. An ultimate parent is classified as an MNE if it has affiliates in more than one country. Based on the ultimate parent firm's country of origin, I then identify foreign MNE affiliates and home MNE affiliates in the BoL data. An ultimate parent with affiliates located in only one country is classified as a non-MNE, meaning that it is a domestic firm appearing in the BoL data. The overall matching and classification process is illustrated in [Figure A1](#).

[Figure 1](#) presents aggregated exports by the multinational status of the firm defined above. As shown in the figure, some firms cannot be classified because the multinational status of either the firm itself or its parent cannot be determined. The matched firms account for approximately 50–80% of total trade flows in each year.¹³ The total aggregated export value from S&P Panjiva appears representative because it closely aligns with the aggregate trade statistics reported in UN Comtrade, indicated by the red dotted line.¹⁴

Exports by MNEs grew in parallel with the overall expansion of India's exports. The sharp increase between 2020 and 2021 likely reflects the recovery in export activity following the COVID-19 pandemic. Sector-level details are provided in [Figure A4](#) and [Figure A5](#). Summarizing the sectoral patterns, the role of MNEs has become particularly important in the Mineral Products and Base Metals sectors. Over the past three years, India's exports of Mineral Products have ranged between USD 60 and 80 billion, with MNEs accounting for more than 80 % of total trade. MNEs also continue to play a substantial role in the Chemical Products, Plastics and Rubber, Machinery and Mechanical Appliances, and Vehicles and Aircraft sectors.

2.1.2 Identifying Related Party and Arm's Length Export

To identify related-party and arm's-length trade, I use two types of string-matching algorithms.¹⁵ The detailed matching procedure is provided in [Appendix A.4](#). During the

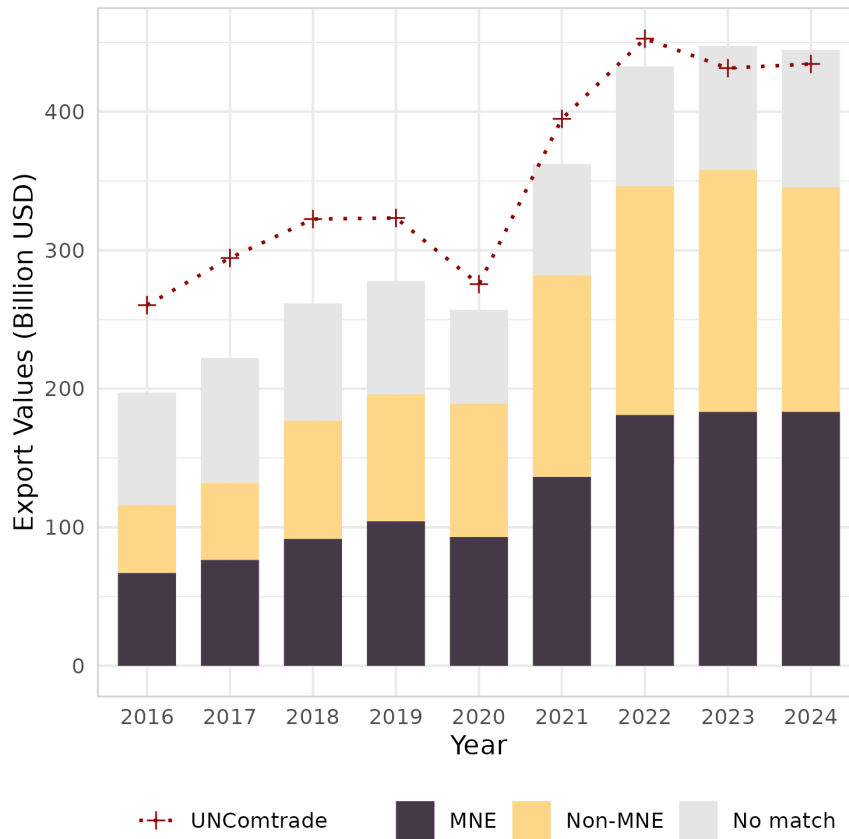
¹²According to S&P's technical documentation, the ultimate parent is defined as "the company at the top of a corporate structure or the legal organization that is ultimately responsible for all associated entities below it" ([S&P Global, 2024](#), p.10).

¹³Observations that are unidentified in terms of MNE status show distinct characteristics, as presented in [Figure A2](#). On average, these observations have substantially smaller trade values, fewer products, and fewer destination markets than firms classified as MNEs. Accordingly, although unmatched observations may cause the measured share of MNE-linked trade to be understated, the magnitude of this understatement may be limited.

¹⁴Nonetheless, some discrepancies remain. These differences are likely attributable to the exclusion of observations with missing destination information during the aggregation process. Although misreporting may also contribute to these discrepancies, investigating such issues lies beyond the scope of this study.

¹⁵One is the Jaccard distance, which measures how different two sets are by comparing the number of shared elements with the total number of unique elements across them. For strings, it typically compares sets of

Figure 1: Export from India by Firm Type



Notes: The y-axis reports export values in billions of U.S. dollars. Multinational enterprise (MNE) status is defined according to the classification of the parent firm. ‘No match’ refers to firms that are either not identified in the S&P firm database or lack parent information. The dark red dashed line represents UN Comtrade export values, which are reported on a free on board (FOB) basis. India’s Bill of Lading export values are also recorded on an FOB basis. The MNE category includes home MNE establishments and foreign MNE affiliates.

matching process, I aim to match firms that either have exactly the same name or share key elements of their names (e.g., Nestlé India and Nestlé U.S.).¹⁶

Technically, if the string-matching process performs well, MNEs should exhibit both related-party (within-firm) and arm’s-length (between-firm) trade. In contrast, non-MNEs should display only between-firm exports. In this study, non-MNEs are defined as firms whose ultimate parent operates solely within a single country. Consequently, it is unlikely that such firms have a parent or subsidiary abroad.

overlapping pieces (called q-grams) taken from each string. Larger values indicate greater similarity. The other is Jaro–Winkler. It measures how similar two strings are and is especially useful for short texts such as names. It evaluates how many characters match and whether they appear in the same order. If the beginning of the two strings is the same, it gives a small bonus to the score. The result is a number between 0 and 1, where 1 means the strings are exactly the same. The Jaro–Winkler distance is 1 minus this score, so smaller values indicate greater similarity. The threshold used to determine related-party trade has been manually tested through trial and error and can be further refined.

¹⁶It is possible that related firms do not share any common name. In the current version of this study, this may lead me to underestimate related-party flows. Future revisions will address this concern.

The matching results in [Figure A3](#) show that the majority of within-firm flows are concentrated in the MNE category, which is the expected and desirable outcome. However, I also observe some within-firm transactions among non-MNEs and unmatched firms. This likely reflects imperfections in identifying firm relationships based on names and firm structure. Nonetheless, given the clear patterns observed among MNEs, I remain reasonably confident in the overall performance of the matching procedure since the rest of the paper focuses only on MNE exports.

2.1.3 Export by MNE Establishments

Finally, I describe related-party and arm's-length exports of MNE establishments in India in [Figure 2](#). The related-party category includes affiliate-to-parent, parent-to-affiliate, and affiliate-to-affiliate shipments. The matching results for India are strong, covering approximately 80–90% of total exports by MNE establishments. While arm's-length exports continue to account for the larger share of MNEs' total exports, related-party exports have expanded rapidly, roughly doubling since 2022. I further find that this increase was driven primarily by the sharp rise in Indian parent MNEs' (home MNEs) related-party exports (that is, parent-to-affiliate intra-firm trade) in [Figure A6](#).

2.2 Other Data

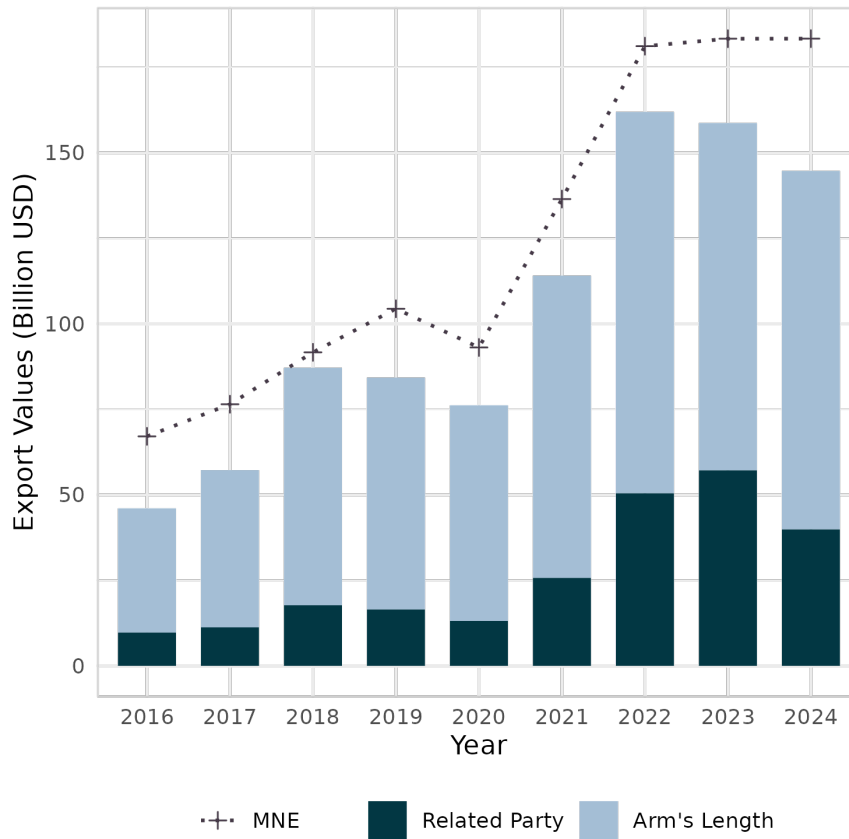
Distance and Other Gravity Controls Distance is obtained from the Centre d'Études Prospectives et d'Informations Internationales (CEPII) Gravity Database, where it is measured as the distance between the largest cities in each country pair. The database also provides additional variables that capture factors influencing bilateral trade costs, such as contiguity, common official language, colonial ties, free trade agreements, common legal origins, and unilateral information on whether a country is currently a WTO member.

GDP and Population GDP and population data are sourced from the World Bank's World Development Indicators (WDI). When data for the most recent year are unavailable, values from the preceding year are used instead.

2.3 Summary

The BoL data used in this study consist of export transactions from India. These data include all firms exporting from India, regardless of whether they are domestic or foreign-owned. Each export record provides identifiers and names for (1) the exporting firm that actually conducts the shipment and (2) the importing firm that receives it. Using

Figure 2: Related Party and Arm's Length Export from MNE Establishment in India



Notes: The y-axis is in billions of dollars. Multinational status is determined by the parent firm's classification. Related-party trade is identified through string matching between exporter and importer names. The dashed line indicates the total exports of MNEs from India illustrated in [Figure 1](#).

supplementary datasets and name-matching procedures, I further identify (1) the parent company of each exporting firm, (2) the multinational status of the parent firm, and (3) whether the exporting and importing firms are related parties.

I take careful steps to minimize potential measurement error in the data processing. First, I exclude Bill of Lading records that lack importer names or destination information, which may lead to a conservative estimate of total trade flows. Second, when matching exporter and importer names, I link firms with identical or closely aligned names, while recognizing that some related entities may not share common keywords in their names. Ongoing work will further refine these matching procedures to enhance data accuracy.

The rest of the paper focuses on MNEs' exports.

3 Facts

This section presents four stylized facts about multinational firms' exports conducted through related-party and arm's-length transactions, with particular attention to how

these patterns vary with distance. Unless otherwise specified, the analysis includes both foreign MNE affiliates operating in India and home MNEs headquartered in India. First, I examine the export patterns of foreign MNEs in Section 3.1. I then explore the relationship between distance and related-party exports in Section 3.2.

3.1 Characteristics of Export Pattern

In my dataset, approximately 2,000 multinational enterprises (parent firms) were operating in India in 2024. The total number of MNEs has increased steadily since 2016 (about 1,700),¹⁷ and this increase is driven by the rise in foreign MNEs. The number of foreign MNEs in India rose from 1,028 in 2016 to 1,190 in 2024. This trend reflects India's growing integration into global production networks and the sustained inflow of foreign direct investment. According to the World Bank, India's net FDI inflows remained consistently positive over the past decade, averaging around 2–3 % of GDP between 2016 and 2019, surging to about 5.4 % in 2020, and stabilizing thereafter at around 0.7 % of GDP (roughly USD 27 billion) in 2024.¹⁸

MNE Trading Partner Composition First, I examine the nature of MNE exports. In particular, I examine whether these MNEs engage primarily in related-party or arm's-length exports. Figure 3 presents the distribution of export modes by MNE type (foreign versus home). About 61–64% of foreign MNEs engage in both related-party and arm's-length exports, indicating that most firms simultaneously serve their own corporate networks and independent buyers. A subset of firms specializes exclusively in one mode, and among these firms, arm's-length exporters outnumber those trading solely within related parties. This pattern suggests that even within multinational networks, firms actively pursue external market opportunities beyond their corporate boundaries. The same feature holds for both foreign and home MNEs, although the share of home MNEs engaging in both export modes is lower than that of foreign MNEs (44–59%). I summarize these findings below in Fact 1:

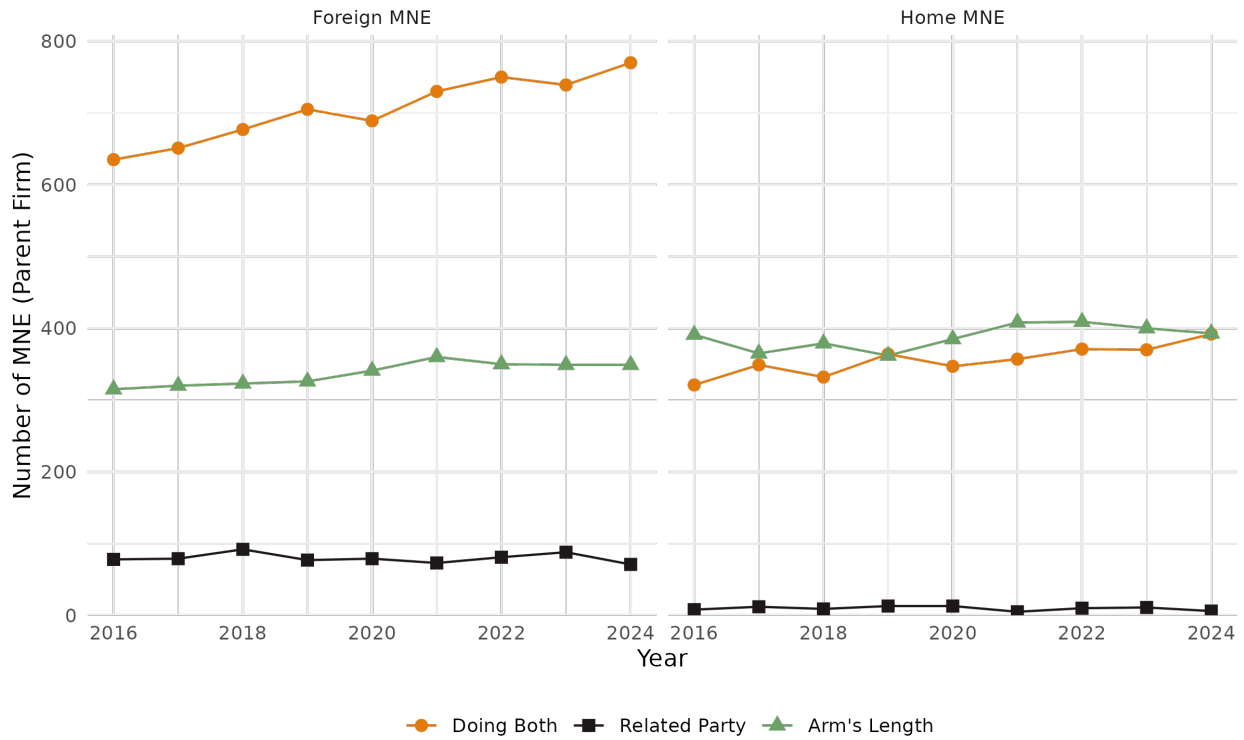
***Fact 1.** MNE affiliate establishments engage in both related-party and arm's-length exports.*

Export Composition Next, I explore the composition of goods that foreign MNEs export through the two channels. In Figure A7, I plot the export values of foreign MNE affiliates by end-use classification defined by the United Nations System of National Accounts (SNA). Exports by firms that engage in both related-party and arm's-length transactions exhibit a similar composition, with both modes dominated by intermediate and capital goods rather

¹⁷Note that this figure refers to parent firms, not affiliates.

¹⁸World Bank, Foreign Direct Investment, Net Inflows (BoP, Current USD), accessed October 2025.

Figure 3: To Whom MNEs Sell



Notes: The y-axis reports the number of parent firms, not affiliate firms, which are the actual exporting entities.

than final goods. However, exports by home MNE affiliates show a markedly different pattern (see [Figure A8](#)). Among home MNEs that export to both related parties and independent buyers, the majority of intermediate goods are sold to independent buyers, while exports to related parties are concentrated in fuels (motor spirits) and passenger motor vehicles. This pattern has been apparent in the data since 2022.

Next, I compare the export structures of the two transaction modes by calculating an export similarity index that quantifies the overlap in export commodity baskets between related-party and arm's-length trade. This allows me to assess the extent to which the goods traded under the two modes are differentiated. The index, originally proposed by [Finger and Kreinin \(1979\)](#) for cross-country comparisons of export composition, is adapted here to the parent-firm level to compare a firm's export structures across transaction modes. Conceptually, the index measures the degree of similarity in a firm's export composition across the two modes. It captures the shared portion of export shares across products—that is, the degree to which the two modes share a common export portfolio. A value of one indicates identical product structures across modes, while a value of zero implies that the firm exports entirely distinct sets of products to related and unrelated buyers. I select firms that engage in both related-party and arm's-length exports. For each firm f and destination j , the index measures how similar the export composition is, in terms of

product mix, between arm's-length and related-party exports. Formally, the measure is defined as follows:

$$ESI_{fj}(\text{Arm}, \text{Related}) = \sum_k \min \left(\frac{\text{Arm}_{fj}^k}{\sum_k \text{Arm}_{fj}^k}, \frac{\text{Related}_{fj}^k}{\sum_k \text{Related}_{fj}^k} \right) \quad (3.1)$$

where k denotes products at the HS 6-digit level. Each term inside the sum compares the share of a particular product k in the firm's total arm's-length exports with the corresponding share in its related-party exports. Taking the minimum of the two shares identifies the overlapping portion of export weight for that product. Summing across all k gives the total overlap in the two distributions. Based on this index, I compute the average share of firms with a value of zero for a given destination. On average, between 94% and 96% of firms exporting to a destination exhibit no overlap in their exported products across the two transaction modes (see Table 1). This finding indicates that most firms export distinct sets of goods under related-party and arm's-length trade to the same destination.

Table 1: Average Percentage (%) of Firms with Zero Export Similarity Between Arm's-Length and Related-Party Exports, by Foreign and Home MNE

	2016	2017	2018	2019	2020	2021	2022	2023	2024
Foreign MNEs	95.47	94.23	94.60	94.90	95.66	95.18	94.54	95.23	95.54
Home MNEs	97.33	97.24	97.45	97.08	97.06	97.16	94.14	93.15	93.69

Note: Author's calculations. Foreign MNEs refer to affiliates of multinational firms headquartered outside India, while Home MNEs refer to Indian multinationals with foreign affiliates.

I summarize these findings below in Fact 2:

Fact 2. *Foreign MNE affiliates mainly sell intermediate goods to both related parties and independent buyers abroad. They typically export distinct sets of products through related-party and arm's-length channels.*

3.2 Related Party Export and Distance

In the following subsection, I examine the relationship between foreign MNE affiliates' related-party exports and geographic distance. I first analyze the binary decision to engage in related-party exports as a function of distance and then explore how the share of related-party exports varies with distance over time.

The Impact of Distance on Probability of Related-Party Export I estimate a linear probability model (LPM) to examine statistically the relationship between distance and

the decision to engage in related-party exports.¹⁹ The dependent variable, Y_{fhijt} , equals 1 if parent firm f , headquartered in h , conducts any *related-party* export from India (i) to destination j in year t , and 0 otherwise. Although the dataset contains all affiliate-level export records, they are aggregated to the parent-firm level for this analysis. Therefore, the unit of analysis is the parent firm rather than the affiliate. Recall that all affiliates are located in India, while their parent firms are headquartered outside India. My main regressor is the log of the distance between India and destination j , $\log(dist_{ij})$. Following the “headquarters-gravity” literature (e.g., [Head and Mayer, 2019](#); [Wang, 2021](#)), I also include the log of the distance between the parent firm’s headquarters h and destination j , $\log(dist_{hj})$. I control for lagged firm-level characteristics measured in India (e.g., the number of destination markets and products exported, and the number of Indian affiliates) to proxy for the scale of production and operations. The model is estimated using ordinary least squares (OLS), and heteroskedasticity-robust standard errors are clustered at the parent-firm level.

$$\begin{aligned}
Pr(Y_{fhijt} = 1 | X_{fhijt}) = & \beta_1 \log(dist_{ij}) + \beta_2 \log(dist_{hj}) + \beta_3 fta_{ijt} \\
& + \beta_4 \log(gdp_{jt}) + \beta_5 \log(pop_{jt}) \\
& + \gamma_F FirmChar_{fi(t-2)} + \\
& + \gamma_{Intra} I(h = j) + \alpha_{ht} + \varepsilon_{fhijt}
\end{aligned} \tag{3.2}$$

On average, I find a positive association between geographic distance and the probability of exporting to a related party, and this relationship is consistent across all model specifications. The effect of distance between the headquarters and destination appears to be negative, although its statistical significance is weaker. Related-party exports to the headquarters country are more likely, which is consistent with previous findings in [Wang \(2021\)](#). I summarize these findings below in Fact 3:

Fact 3. *On average, there is a positive association between geographic distance and the probability of exporting to a related party.*

The Evolution of Related-Party Export [Figure 4](#) illustrates the share of related-party trade across distance bins. Distance refers to the great-circle distance between the most populous cities of each country, as provided by the CEPII gravity database. The distances are sorted from shortest to longest and divided into ten equal-count bins, so that each bin contains roughly the same number of countries. These bins do not represent equal distance intervals, but they still capture the ordering of distances. The distance bins correspond

¹⁹It is well known that the LPM is straightforward to estimate and easy to interpret, though it may not be the most appropriate specification for modeling binary outcomes because of its linearity and the potential for predicted probabilities to fall outside the [0,1] range. Nevertheless, when the primary interest lies in estimating the average partial effect, the LPM can yield meaningful results under certain conditions, as discussed in [Chen et al. \(2023\)](#).

Table 2: Probability of Related-Party Trade by Foreign MNEs in India: OLS Estimates

	Pr(Related party export to j in $t = 1$)					
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(dist_{ij})$	0.073*** (0.005)	0.062*** (0.006)	0.058*** (0.007)	0.063*** (0.005)	0.061*** (0.006)	0.056*** (0.007)
$\log(dist_{hj})$	-0.011* (0.004)	-0.007 (0.005)	-0.007 (0.006)	-0.007* (0.004)	-0.006 (0.004)	-0.006 (0.005)
fta_{ijt}	0.095*** (0.009)	0.102*** (0.009)	0.098*** (0.009)	0.071*** (0.008)	0.072*** (0.009)	0.068*** (0.009)
$\log(gdp_{jt})$	0.087*** (0.003)	0.095*** (0.004)	0.096*** (0.004)	0.089*** (0.004)	0.091*** (0.004)	0.091*** (0.004)
$\log(pop_{jt})$	-0.031*** (0.003)	-0.031*** (0.003)	-0.029*** (0.003)	-0.029*** (0.003)	-0.028*** (0.003)	-0.026*** (0.003)
wto_{jt}	0.012 (0.010)	0.014 (0.011)	0.011 (0.011)	0.022* (0.009)	0.021* (0.010)	0.017+ (0.010)
$I(h = j)$	0.170*** (0.018)	0.196*** (0.019)	0.191*** (0.023)	0.192*** (0.015)	0.198*** (0.017)	0.185*** (0.021)
number of market (t-2)		0.000 (0.000)	0.000 (0.000)			
number of goods (t-2)		0.001*** (0.000)	0.001*** (0.000)			
number of affiliates (t-2)		0.006 (0.006)	0.006 (0.006)			
R ²	0.164	0.185	0.185	0.403	0.396	0.397
Observations	134,429	97,525	97,177	134,429	97,525	97,177
Other Gravity	N	N	Y	N	N	Y
Headquarter Country - Year FE	✓	✓	✓			
Parent Firm (unit) - Year FE				✓	✓	✓

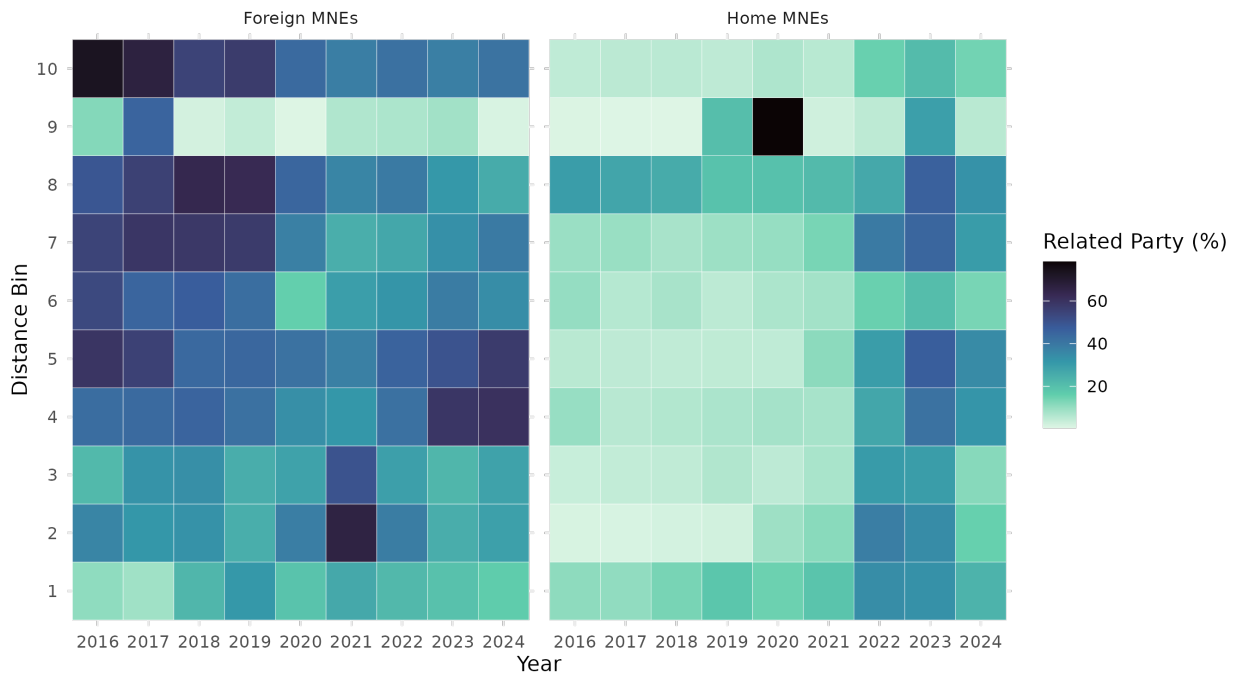
Signif. Codes: ***: 0.001, **: 0.01, *: 0.05. The unit of regression is the parent firm (headquarters), and standard errors are clustered at this level. $\log(dist_{ij})$ denotes the distance between the headquarters of a foreign MNE and destination j . $\log(dist_{hj})$ represents the distance between India and destination j . fta_{ijt} is a binary variable indicating whether a Free Trade Agreement exists between India and j at time t . $\log(gdp_{jt})$ and $\log(pop_{jt})$ denote the GDP and population of destination j , respectively. wto_{jt} equals 1 if destination j is a WTO member, and 0 otherwise. $I(h = j)$ is an indicator variable that equals 1 if the destination of export is the same as the headquarters country of the foreign MNE. *Other Gravity* variables include bilateral characteristics defined between India and destination j , as well as between the headquarters country h and destination j . These include colonial relationship, common language, contiguity, and shared legal origins.

to deciles of the distribution, with detailed country lists provided in [Table A2](#). Export values are aggregated into the ten distance bins, and the share of related-party exports is calculated for each bin.

Foreign MNE affiliates exhibited a relatively clear pattern in their related-party exports: a larger share of such trade occurred with geographically distant destinations, suggesting a positive association between distance and the intensity of related-party transactions. This pattern persisted until 2020 but has gradually weakened since then. The data suggest that this change is driven by an increase in arm's-length exports rather than a decline in related-party exports ([Figure A6](#)). The same pattern holds when the related-party share is computed at the affiliate level and weighted by each affiliate's total exports to each bin ([Figure A9](#)). I summarize these findings below in Fact 4:

Fact 4. *Until 2020, arm's-length exports were more common at shorter distances, whereas related-party exports dominated at longer distances. However, this pattern has weakened in recent years.*

Figure 4: Related Party Trade Share by Distance Bin and Year - India



Notes: This figure illustrates the related-party trade share by distance for MNE firms headquartered outside of India versus those headquartered in India. The ratio is calculated at each bin using aggregated trade values. This analysis includes multinational firms only. For India, the seventh bin includes West and Southern African countries such as South Africa and Nigeria. The eighth bin covers Canada and United States, Oceania and some West African countries.

4 Theoretical Framework

In this section, I construct a partial-equilibrium model of related-party and arm's-length exports that captures the stylized facts and provides the foundation for the estimating equations. The framework is a firm-level, multi-sector adaptation of [Eaton and Kortum \(2002\)](#), featuring a representative downstream buyer and multiple suppliers located across countries. I assume that all firms in the model are affiliates of MNEs. Production takes place in multiple sectors, indexed by $g \in \mathcal{G}$. In each destination country $j \in J$, a representative buyer in sector g produces and sells a fixed quantity of a final good using imported inputs. The final-good production function is Cobb–Douglas across sectors, while within each sector, inputs from different suppliers are aggregated using a constant elasticity of substitution (CES) function with elasticity of substitution $\sigma_g > 1$. The buyer minimizes total input costs by sourcing varieties from multiple origin countries $n \in N$ and from a large set of potential suppliers $i \in S_n$ within each origin. Each supplier produces a continuum of differentiated varieties ω and can serve the buyer through one of two channels (modes): *related-party* (within-firm) or *arm's-length* (between-firm) exports. For a given supplier, the sets of varieties exported under each mode are distinct, reflecting mode-specific productivity draws and trade costs.

4.1 Technology

Suppliers in sector g draw efficiency at the variety level from a common technology frontier that does not vary across sectors. This specification implies that the distribution of draws is invariant across sectors g , so the sector index does not enter the technology parameters. Specifically, each supplier i in source country n draws idiosyncratic productivity terms $z_{ni}^m(\omega)$ independently across varieties $\omega \in \Omega_g$ and modes $m \in \{W, B\}$ from a Fréchet distribution with country–supplier location parameter $T_{ni} > 0$ and common shape parameter $\theta > 1$:

$$\Pr(z_{ni}^m(\omega) \leq z) = \exp\left[-T_{ni} z^{-\theta}\right].$$

This implies that suppliers' relative efficiency across modes is random across varieties. A supplier may be relatively more efficient at producing certain varieties for related-party exports and others for arm's-length exports, even though both arise from the same underlying technology shared across sectors.

Let c_{ng} denote the unit production cost in source country n and sector g . The delivered price of variety ω from supplier i in sector g and source country n to destination j via mode m is therefore

$$p_{nij}^{g,m}(\omega) = \frac{c_n^g}{z_{ni}^m(\omega)} \tau_{nj}^{g,m},$$

where $\tau_{nj}^{g,m}$ represents the iceberg trade cost between n and j in sector g under mode m .

To summarize, technology parameters are common across sectors ($\theta_g = \theta$), which implies that the statistical dispersion of variety-level productivity draws for firm i is not sector-specific. Supply-side sectoral differences arise from sector-level unit production costs and trade frictions, not from differences in the underlying degree of technological dispersion.

4.2 Demand

Buyers are final-good producers, and they use a two-tier input aggregation structure for production. Final producers in j aggregate inputs across sectors with Cobb–Douglas shares $\alpha_g > 0$ (tier 1) and combine a continuum of inputs $\omega \in \Omega_g$ using a CES aggregator with elasticity $\sigma_g > 1$ within each sector (tier 2).

Final output in j is Cobb–Douglas across sectors:

$$Q_j = \prod_{g \in \mathcal{G}} Q_j^g{}^{\alpha_g}, \quad Q_j^g = \left(\int_{\Omega_g} q_j^g(\omega)^{\frac{\sigma_g-1}{\sigma_g}} d\omega \right)^{\frac{\sigma_g}{\sigma_g-1}}.$$

The final-good producer in j sells an exogenous quantity of the final product and cost-minimizes over a continuum of intermediate inputs sourced from multiple suppliers, either at arm's length or from affiliates.

4.3 Equilibrium

The equilibrium is characterized by prices, supplier sourcing probabilities that correspond to destination expenditure shares, and the firm-to-destination gravity equation. Detailed derivations are provided in Appendix B.3.

4.3.1 Price

Supplier side price offer. Denote by $G_{nij}^m(p)$ the probability that supplier i in n can offer buyer j variety ω through export channel (mode) m at a price below p .

$$\begin{aligned} G_{nij}^{g,m}(p) &\equiv \Pr\left(p_{nij}^{g,m}(\omega) \leq p\right) = \Pr\left(\frac{c_n^g}{z_{ni}^m(\omega)} \tau_{nj}^{g,m} \leq p\right) = \Pr\left(z_{ni}^m(\omega) \geq \frac{c_n^g \tau_{nj}^{g,m}}{p}\right) \\ &= 1 - \exp\left\{-T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta} p^\theta\right\}. \end{aligned}$$

Cheapest price distribution that is accepted by buyer The lowest price is the realization of the minimum delivered price among all supplier–mode pairs available to buyers in j , which can be written as $p_j^g(\omega) \equiv \min_{n,i,m} \{p_{nij}^{g,m}(\omega)\}$. In other words, the buyer chooses the cheapest option across modes and source countries. Assuming all prices are independent and all buyers in j are identical, I can derive the distribution of prices across goods for buyers in country j .

$$\begin{aligned} G_j^g(p) &= \Pr(p_j^g(\omega) \leq p) \\ &= \Pr\left(\min_{ni,m} \{p_{nij}^{g,m}(\omega)\} \leq p\right) \\ &= 1 - \exp\left\{-\Phi_j^g p^\theta\right\} \quad \text{where } \Phi_j^g \equiv \sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta} \end{aligned}$$

4.3.2 Total expenditure in j on a supplier's channel

Under a CES demand system, I derive Marshallian input demand (see Appendix B.1) with price index $P_j^g = \left(\int_{\Omega_g} p_j^g(\omega)^{1-\sigma_g} d\omega\right)^{\frac{1}{1-\sigma_g}}$

$$q_j^g(\omega) = \frac{E_j^g}{P_j^g} \left(\frac{p_j^g(\omega)}{P_j^g}\right)^{-\sigma_g} \quad (4.1)$$

Integrating the demand in equation (4.1) over the set of varieties sourced from supplier i in country n via channel (mode) m yields total expenditure

$$X_{nij}^{g,m} = \int_{\omega \in \Omega_{nij}^{g,m}} p_j^g(\omega) q_j^g(\omega) d\omega = E_j^g \cdot \underbrace{\Pr(p_{nij}^{g,m}(\omega) = p_j^g(\omega))}_{\text{sourcing probability}}.$$

4.3.3 Sourcing probability

Next, I derive the probability that supplier i in source country n , using mode m , is the lowest-cost provider of variety ω in destination j . I refer to this probability as the *sourcing probability*. It represents the likelihood that supplier i in source country n , using mode m , offers the minimum delivered price among all potential supplier–channel pairs. Recall that firm i has two channels through which it can export goods, denoted by modes $m \in \{W, B\}$, where W represents exports to related parties (Within-firm) and B represents exports to independent buyers (Between-firm, or arm's-length, trade).

Because there is a continuum of goods and each good receives an independent and identically distributed mode-specific productivity draw, the law of large numbers implies that the share of varieties for which supplier i in source country n , using mode m , is the

lowest-cost provider equals the probability of being the lowest-cost supplier for any given variety.

The fraction of goods in sector g that firm i exports to destination j using mode $m = W$ corresponds to the share of its exports conducted through related-party transactions.

$$\begin{aligned}\Pi_{nij}^{g,W} &\equiv \Pr\left(p_{nij}^{g,W}(\omega) \leq \min_{(n'i',m') \neq (ni,W)} p_{n'i'j}^{g,m'}(\omega)\right) \\ &= \frac{T_{ni}(c_n^g \tau_{nj}^{g,W})^{-\theta}}{\sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} T_{ni}(c_n^g \tau_{nj}^{g,m})^{-\theta}}\end{aligned}\quad (4.2)$$

The same logic applies to the case $m = B$. Hence, the fraction of goods that firm i in n exports to j using mode (channel) $m = B$ is:

$$\Pi_{nij}^{g,B} = \frac{T_{ni}(c_n^g \tau_{nj}^{g,B})^{-\theta}}{\sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} T_{ni}(c_n^g \tau_{nj}^{g,m})^{-\theta}}\quad (4.3)$$

4.3.4 Trade Flow by Channel (Firm-to-Destination)

Finally, the expenditure share of country j on supplier i via channel (mode) m equals its sourcing probability:

$$\frac{X_{nij}^{g,m}}{E_j^g} = \Pi_{nij}^{g,m}\quad (4.4)$$

where $m \in \{W, B\}$ and E_j^g is total expenditure by buyer j in sector g .

This yields the expenditure in j on goods sold by firm i in source country n :

$$X_{nij}^{g,m} = E_j^g \Pi_{nij}^{g,m} = E_j^g \frac{T_{ni}(c_n^g \tau_{nj}^{g,m})^{-\theta}}{\Phi_j^g}.$$

Substituting $\Phi_j^g = P_j^{g-\theta} C^{g\theta}$ gives

$$X_{nij}^{g,m} = E_j^g T_{ni} \left(p_j^{g-1} c_n^g \tau_{nj}^{g,m} \right)^{-\theta} C^{g\theta}\quad (4.5)$$

where P_j is the price index in the destination market derived in [Eaton and Kortum \(2002\)](#), and $C^g = \Gamma\left(\frac{\theta+1-\sigma_g}{\theta}\right)^{\frac{1}{1-\sigma_g}}$ is a sector-specific constant.

4.4 Trade Costs

I parameterize iceberg costs by mode $m \in \{W, B\}$ as

$$\tau_{nj}^{g,m} = o_n^g \iota_j^g d_{nj}^{\delta_m} \exp\{z'_{nj} \Gamma_m\}, \quad (4.6)$$

where o_n^g and ι_j^g capture origin- and destination-level trade cost shifters, d_{nj} is the bilateral distance between source n and destination j , and δ_m is the mode-specific distance elasticity of trade costs. The vector z_{nj} collects other bilateral factors (e.g., contiguity, common language, or FTA), and Γ_m is a conformable vector of mode-specific loadings. This specification ensures $\tau_{nj}^{g,m} > 0$ and allows dummy or zero-valued covariates to enter linearly within the exponential.

4.5 Comparative Statics

Substituting (4.6) into the sourcing probabilities (4.3) and (4.4), the ratio of within- to between-firm exports becomes

Define the aggregate within/arm's-length ratio

$$R_{nij}^{\text{agg}} \equiv \frac{\sum_g X_{nijg}^W}{\sum_g X_{nijg}^B} = \frac{\sum_g B_{jg} d_{nj}^{-\theta \delta_W} \exp\{-\theta z'_{nj} \Gamma_W\}}{\sum_g B_{jg} d_{nj}^{-\theta \delta_B} \exp\{-\theta z'_{nj} \Gamma_B\}},$$

where

$$B_{jg} \equiv \alpha_g E_j \frac{T_{ni} (c_{ng} o_{ng})^{-\theta}}{\tilde{\Phi}_{jg}},$$

and

$$\tilde{\Phi}_{jg} = \sum_{n',i',m'} T_{n'i'} (c_{n'g} o_{n'g})^{-\theta} d_{n'j}^{-\theta \delta_{m'}} \exp\{-\theta z'_{n'j} \Gamma_{m'}\}.$$

Because B_{jg} does not depend on m , it cancels across the sums:

$$R_{nij}^{\text{agg}} = d_{nj}^{-\theta(\delta_W - \delta_B)} \exp\{-\theta z'_{nj} (\Gamma_W - \Gamma_B)\}.$$

Taking logs and differentiating w.r.t. $\ln d_{nj}$ yields:

$$\frac{\partial \ln R_{nij}^{\text{agg}}}{\partial \ln d_{nj}} = -\theta(\delta_W - \delta_B).$$

Implication. If related-party (within-firm) exports are *less* sensitive to distance than arm's-length exports, so that $\delta_W < \delta_B$, then $\frac{\partial \ln R_{nij}^{\text{agg}}}{\partial \ln d_{nj}} > 0$. In this case, the ratio $\frac{X_{nij}^{W,\text{agg}}}{X_{nij}^{B,\text{agg}}}$

increases with distance, consistent with the stylized fact that related-party exports tended to dominate at longer distances during 2016–2020. If the difference in distance elasticities between the two channels ($\delta_W - \delta_B$) has declined over time, this could help explain the attenuation of that pattern observed after 2020.

5 Empirical specification

Deriving an estimable equation. Substituting the trade-cost parameterization in (4.6) into the firm-level gravity equation in (4.5) gives, for each sector g and mode $m \in \{W, B\}$,

$$X_{nij}^{g,m} = E_j^g T_{ni} \left((P_j^g)^{-1} c_n^g o_n^g \iota_j^g d_{nj}^{\delta_m} \exp\{Z'_{nj} \Gamma_m\} \right)^{-\theta} (C^g)^{-\theta}. \quad (5.1)$$

Equation (5.1) is an exact structural relationship. For empirical implementation, I interpret it as a restriction on the conditional mean. Let L denote the collection of observables on the right-hand side of (5.1). Then

$$\begin{aligned} \mu_{nij}^{g,m} &\equiv E[X_{nij}^{g,m} \mid L, m] \\ &= E_j^g T_{ni} \left((P_j^g)^{-1} c_n^g o_n^g \iota_j^g d_{nj}^{\delta_m} \exp\{Z'_{nj} \Gamma_m\} \right)^{-\theta} (C^g)^{-\theta}. \end{aligned} \quad (5.2)$$

Rewriting (5.2) in exponential-index form gives:

$$\begin{aligned} \mu_{nij}^{g,m} = \exp \left\{ \underbrace{\ln E_j^g + \theta \ln P_j^g - \theta \ln \iota_j^g}_{\text{destination-sector component}} + \underbrace{\ln T_{ni}}_{\text{supplier component}} \right. \\ \left. + \underbrace{-\theta \ln c_n^g - \theta \ln o_n^g - \theta \ln C^g}_{\text{origin-sector component}} \right. \\ \left. - \theta \delta_m \ln d_{nj} - \theta \sum_{z \in \mathcal{Z}} \gamma_{m,z} Z_{nj,z} \right\}. \end{aligned} \quad (5.3)$$

To simplify notation, define

$$\begin{aligned} \varsigma_j^g &\equiv \ln E_j^g + \theta \ln P_j^g - \theta \ln \iota_j^g, \\ \psi_i &\equiv \ln T_{ni}, \\ \eta_n^g &\equiv -\theta \ln c_n^g - \theta \ln o_n^g - \theta \ln C^g. \end{aligned}$$

Then the mode-specific conditional mean becomes

$$\mu_{nij}^{g,m} = \exp \left\{ \zeta_j^g + \psi_i + \eta_n^g - \theta \delta_m \ln d_{nj} - \theta \sum_{z \in \mathcal{Z}} \gamma_{m,z} z_{nj,z} \right\}.$$

Equation (5) shows that the model implies two conditional mean equations, one for related-party trade ($m = W$) and one for arm's-length trade ($m = B$). Since the transaction mode is observed in the data, these two equations can be estimated jointly in a pooled specification.

Pooled specification with a mode indicator. Let $\ell = 1\{m = W\}$, so that $\ell = 1$ for related-party trade and $\ell = 0$ for arm's-length trade. To illustrate the pooling argument compactly, consider the stylized mode-specific conditional means

$$\mu_W(x) = \exp(x' \beta_W), \quad \mu_B(x) = \exp(x' \beta_B).$$

Because the mode is observed, the two conditional means can be written jointly as

$$\mu(x, \ell) = \mu_W(x)^\ell \mu_B(x)^{1-\ell}.$$

Taking logs gives

$$\begin{aligned} \ln \mu(x, \ell) &= \ell x' \beta_W + (1 - \ell) x' \beta_B \\ &= x' \beta_B + \ell x' (\beta_W - \beta_B), \end{aligned} \tag{5.4}$$

and hence

$$\mu(x, \ell) = \exp(x' \beta_B + \ell x' (\beta_W - \beta_B)). \tag{5.5}$$

Thus the pooled model is a reparameterization of the two mode-specific conditional means, with the coefficients on the interaction terms capturing the difference between related-party and arm's-length trade.

Applying this reparameterization to (5), and taking arm's-length trade ($m = B$) as the baseline, yields

$$E[X_{nijm}^g | L, m] = \exp \left\{ \zeta_j^g + \psi_i + \eta_n^g + \underbrace{\beta_d^\Delta (\ln d_{nj}) \cdot 1\{m = W\}}_{\text{W-B difference}} + \underbrace{\beta_d^B \ln d_{nj}}_{\text{baseline (B)}} + \sum_{z \in \mathcal{Z}} \left[\beta_z^\Delta (z_{nj,z} \cdot 1\{m = W\}) + \beta_z^B z_{nj,z} \right] + \beta_0 1\{m = W\} \right\}. \quad (5.6)$$

The coefficient mappings are

$$\begin{aligned} \beta_d^B &= -\theta \delta_B, & \beta_d^\Delta &= -\theta(\delta_W - \delta_B), \\ \beta_z^B &= -\theta \gamma_{B,z}, & \beta_z^\Delta &= -\theta(\gamma_{W,z} - \gamma_{B,z}). \end{aligned} \quad (5.7)$$

Therefore, β_d^B captures the distance elasticity for arm's-length trade, while β_d^Δ captures the difference between the distance elasticities of related-party and arm's-length trade. A positive estimate of β_d^Δ implies that related-party exports are less sensitive to distance than arm's-length exports.

If trade costs differ across modes only through distance and the bilateral covariates $z_{nj,z}$, then the model implies $\beta_0 = 0$. In the empirical specification, however, I allow for a mode-specific intercept to absorb residual average differences between related-party and arm's-length trade that are not captured by the observed trade-cost proxies.

To summarize, identification comes from within-firm variation across destinations for the same product, after controlling for supplier, origin-sector, and destination-sector components. The interaction between distance and the related-party indicator captures the difference in distance sensitivity between arm's-length and within-firm transactions.

5.1 Estimation

Following [Gourieroux et al. \(1984b,a\)](#); [Wooldridge \(1999\)](#); [Santos Silva and Tenreyro \(2006, 2011\)](#), I estimate the parameter vector β in (5.7) using Poisson pseudo-maximum likelihood (PPML)²⁰ and interpret the model as a restriction on the conditional mean. Let

$$\mu_{nijm}^g(x; \beta) \equiv E[X_{nijm}^g | L, m]$$

²⁰Poisson regression uses the likelihood function of the Poisson model but does not require the conditional variance to equal the conditional mean. The Poisson pseudo-maximum likelihood (PPML) estimator is the maximum-likelihood estimator based on this conditional mean specification. Because its first-order condition coincides with the sample moment condition in (5.8), the estimator that solves that condition is referred to as the PPML estimator.

be the conditional mean given by (5.6). In pseudo- or quasi-maximum-likelihood estimation (Gourieroux et al., 1984b), the estimator $\hat{\beta}$ is obtained by solving the sample moment conditions

$$\sum_{n,i,j,m} x_{nijm}^g \left[X_{nijm}^g - \mu_{nijm}^g(x; \beta) \right] = 0. \quad (5.8)$$

Because the vector β is common across sectors in the model, I pool all sectors (HS 6-digit products) and absorb sectoral heterogeneity using origin–sector and destination–sector fixed effects. I report robust multi-way clustered standard errors, which remain valid under general forms of heteroskedasticity. PPML also accommodates zero trade flows naturally (Santos Silva and Tenreyro, 2006; Wooldridge, 2023), including settings in which zeros are prevalent (Santos Silva and Tenreyro, 2011).

The regression dataset includes the top 60 destination countries for India and Mexico from 2016 to 2024. Mexico is included at the estimation stage to provide cross-origin variation, which allows destination–product fixed effects to be absorbed. The Mexican data are therefore used for identification purposes. Destination countries are selected annually, and all MNE affiliates that export to these destinations are included in the data. Restricting the sample to major destination countries helps address the lumpiness of the data. The dataset consists of unique firm–product pairs across all selected destinations. If a firm–product pair (i.e., a firm exporting a specific HS 6-digit product) is observed for some destinations, zero-export observations are included for other destinations. Due to computational constraints associated with the nonlinear estimation, regressions are estimated separately for each year. Therefore, any shocks common to all observations within a year are constant and do not affect identification.

6 Results

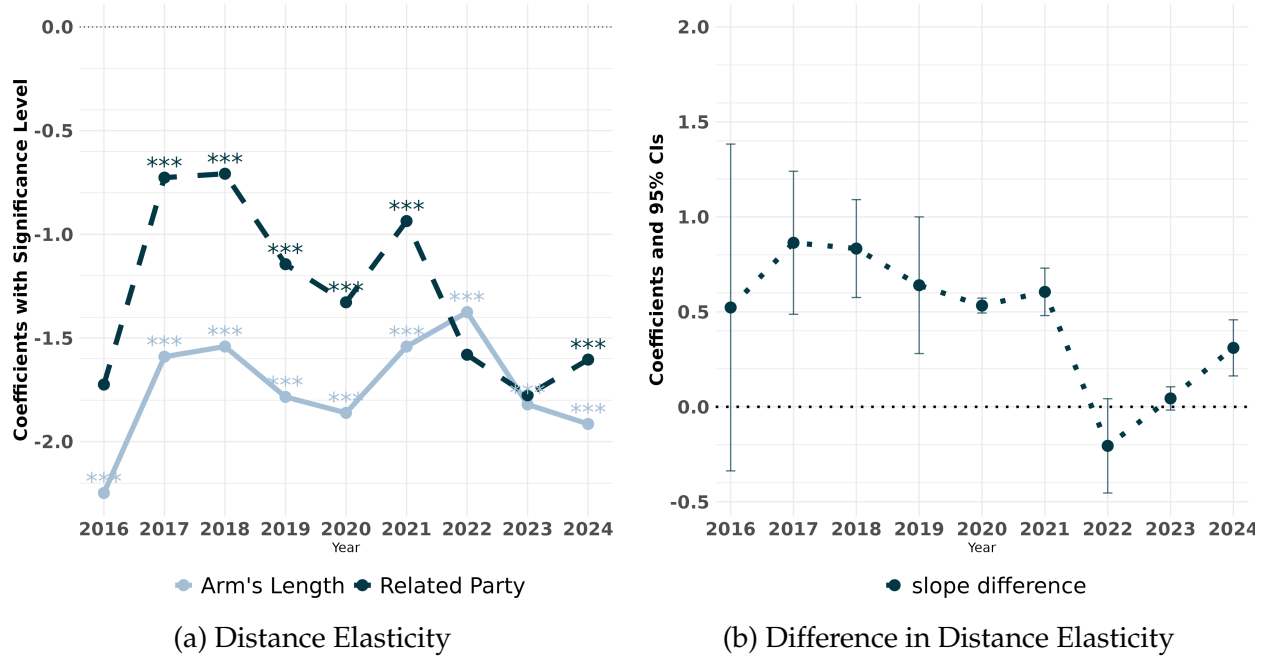
6.1 Baseline

The PPML estimates indicate that related-party exports are less elastic to distance than arm’s-length exports, but this difference has eroded in recent years. Figure 5 presents the results. The left panel reports the coefficients $\beta_d^B = -\theta\delta_B$ and $\beta_d^B + \beta_d^\Delta = -\theta\delta_W$, while the right panel shows $\beta_d^\Delta = -\theta(\delta_W - \delta_B)$, which captures the difference in distance elasticities between the two export modes. Because distance elasticities are negative, a positive value of β_d^Δ implies that related-party exports are less elastic (or less sensitive) to distance than arm’s-length exports.

Between 2016 and 2021, across-firm distance elasticities ($-\theta\delta_B$) remain strongly negative, generally between -1.5 and -2.3 , while within-firm elasticities are consistently much smaller in magnitude. In every year prior to 2022, within-firm trade shows substantially

lower distance sensitivity—for example, -0.73 vs. -1.59 in 2017 and -0.94 vs. -1.54 in 2021—indicating a persistent and sizable within-firm distance-cost advantage throughout this period. Starting in 2022, however, this gap begins to shrink (-1.58 vs. -1.38 in 2022; -1.78 vs. -1.82 in 2023), and the two elasticities are no longer statistically different. A modest rebound appears in 2024, but overall the pattern points to an erosion of the within-firm distance-cost advantage in recent years.

Figure 5: Heterogeneous distance elasticity: related-party vs. arm's-length export



Notes: Robust standard errors are multi-way clustered by MNE affiliate, source country, product, and destination to account for correlated errors across multiple dimensions. The number of observations used in the regressions by year is as follows: 2016 – 6,129,173; 2017 – 6,406,351; 2018 – 7,326,832; 2019 – 7,264,076; 2020 – 6,774,926; 2021 – 7,378,666; 2022 – 7,430,890; 2023 – 7,680,547; and 2024 – 8,516,207.

To place these estimates in context, existing gravity estimates vary with the level of aggregation and the estimation method. Country-level estimates are typically around -1 (Head and Mayer, 2014), while industry-level estimates are often somewhat larger in magnitude, centered around -1.4 (Berthelon and Freund, 2008). At the firm level, Bas et al. (2017) report an elasticity of roughly -0.5 using data on large exporting firms in France and China.²¹ My estimates show that the elasticity is substantially smaller for within-firm exports than for arm's-length exports through 2021. The within-firm estimates, which range from about -0.5 to -1 , are close to the firm-level magnitudes reported by Bas et al. (2017). In contrast, the arm's-length estimates, which range from about -1 to -2 , are closer to the industry-level magnitudes. One possible interpretation is that firm-level

²¹The focus of Bas et al. (2017) is trade elasticities rather than distance elasticities. Distance elasticities are estimated as a part of gravity controls. They use a different estimating equation and estimator from ours, so comparisons should be made with caution.

estimates such as [Bas et al. \(2017\)](#) average across transactions that differ in their sensitivity to distance.

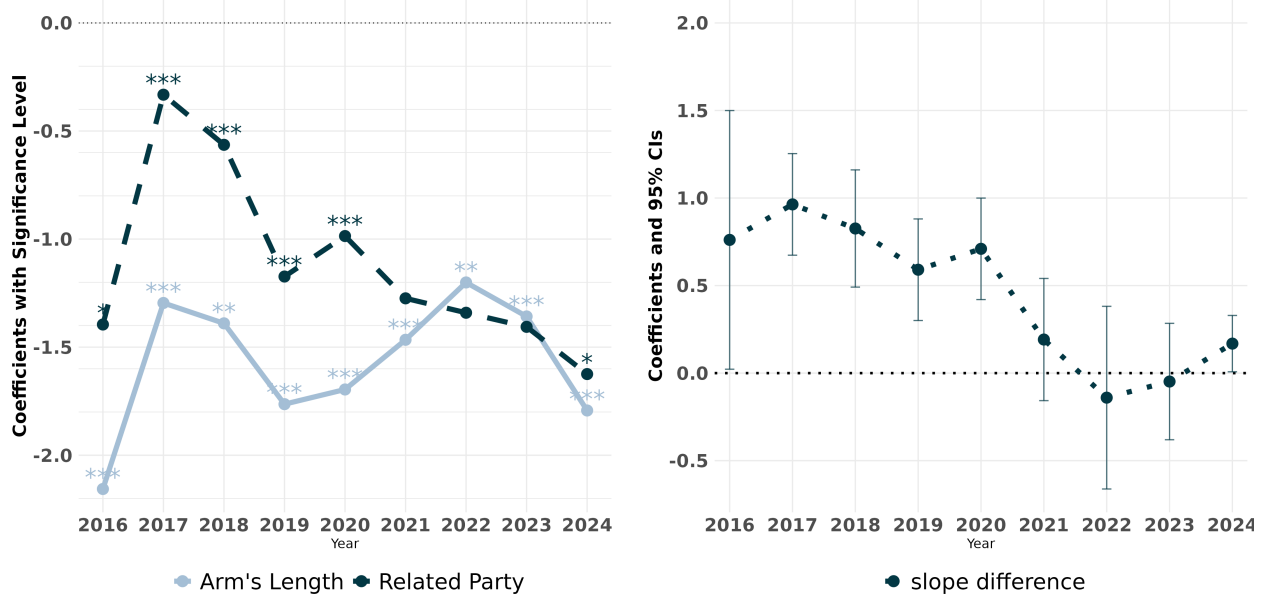
Other Gravity Controls [Figure A10](#) presents the estimated effects of additional proxies for trade costs. I find that the presence of a Free Trade Agreement (FTA) between countries has a positive effect on related-party trade, although the statistical significance varies and is stronger before 2020. This pattern may reflect the fact that relatively few new FTAs have been established in recent years. Interestingly, the effect of FTAs on arm's-length trade is not statistically different from zero in any year. A shared official language is positively associated with related-party trade, while historical colonial ties appear to be more relevant for arm's-length trade, though these effects are statistically significant only in the two most recent years. Finally, while geographic contiguity has a negative effect on arm's-length trade, it is positively associated with related-party trade. Overall, gravity variables other than distance exert limited influence on arm's-length exports, whereas several continue to play a meaningful role in shaping related-party exports.

6.2 Robustness to Product Composition and Firm Dynamics

Product Composition I re-estimate the model on a restricted sample that includes only products that firms export through both transaction modes. This reduces the concern that the estimated mode-specific distance elasticities are driven by product composition rather than differences in transaction mode. For example, if firms systematically use related-party trade for products that are more standardized, higher value, or better suited to long-distance shipping, then part of the difference in distance elasticities may reflect product selection rather than transaction mode. The results are presented in [Figure 6](#). They remain similar to the baseline estimates, with narrower confidence intervals. This suggests that the baseline results are unlikely to be driven by compositional differences in products across transaction modes. The evolution of the gap between the distance elasticities of related-party and arm's-length trade also closely mirrors the baseline pattern, although the shift now appears to begin in 2021 rather than 2022. The same pattern also holds in a further restricted sample consisting of products exported through both transaction modes and continuously exported throughout the sample period ([Figure 7](#)).

Firm Dynamics Since the model is estimated separately by year, neither new MNE entry into India nor changes in the export behavior of incumbent MNEs are explicitly controlled for in the baseline regression. This raises the possibility that changes in the firm sample contribute to the observed shift. To assess whether sample composition drives the result, I re-estimate the model using only MNEs observed in all nine years from 2016 to 2024. The results are consistent with the baseline, but they show more clearly that the

Figure 6: Robustness to Year-Specific Product Samples

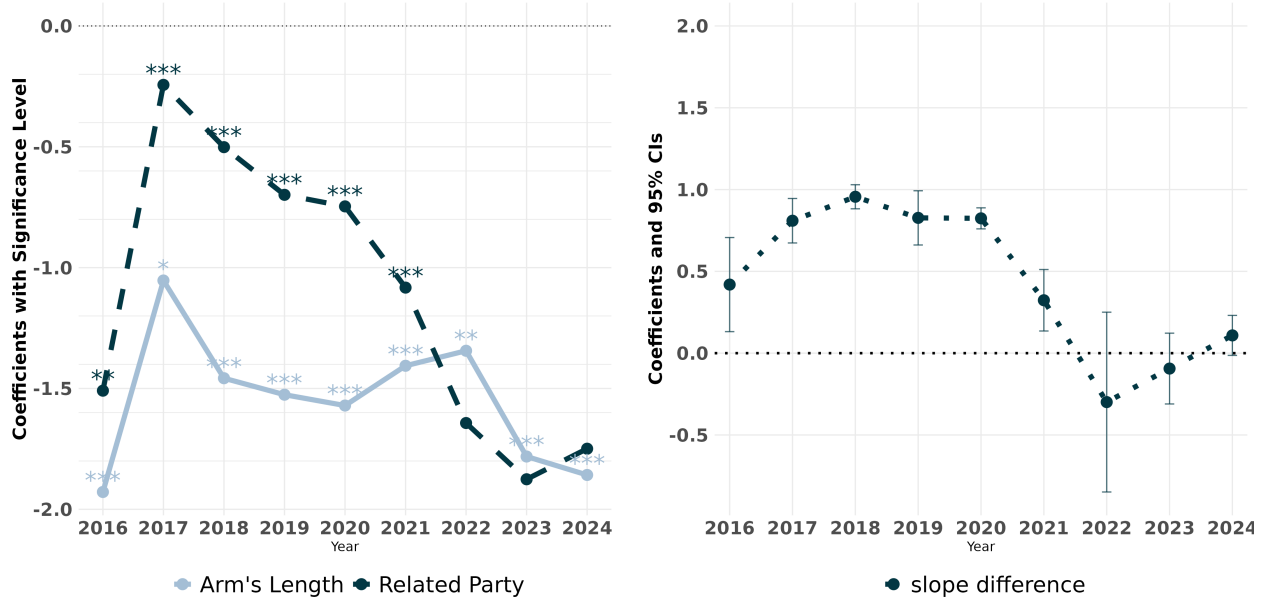


(a) Distance Elasticity

(b) Difference in Distance Elasticity

Notes: Results from a restricted sample of products exported through both transaction modes (within-party and arm's length). Robust standard errors are multi-way clustered by MNE affiliate, source country, product, and destination to account for correlated errors across multiple dimensions. The number of observations used in the regressions by year is as follows: 2016 – 793,920; 2017 – 912,617; 2018 – 1,067,268; 2019 – 1,154,680; 2020 – 1,033,806; 2021 – 1,218,608; 2022 – 1,242,108; 2023 – 1,332,140; and 2024 – 1,478,489.

Figure 7: Robustness to Balanced Product Samples Across Years



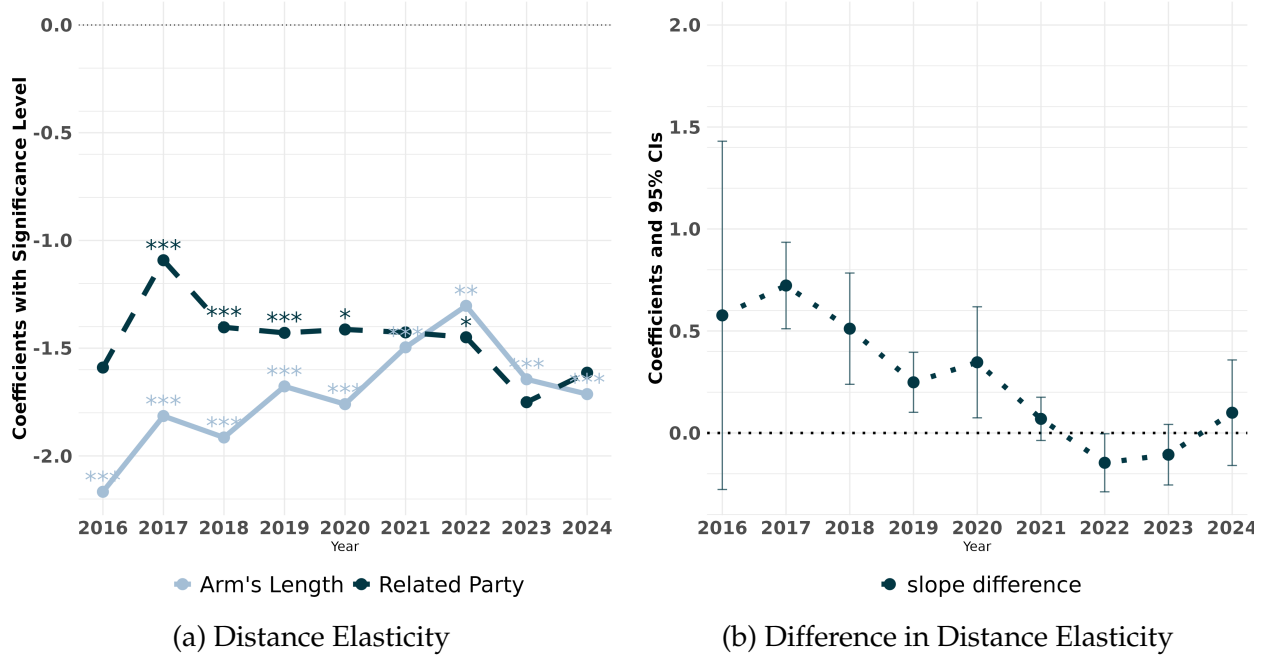
(a) Distance Elasticity

(b) Difference in Distance Elasticity

Notes: Results from a restricted sample of products that are exported through both transaction modes (within-party and arm's length) and are continuously exported throughout the sample period. The total number of unique HS 6-digit products in the sample is 1,677. Robust standard errors are multi-way clustered by MNE affiliate, source country, product, and destination to account for correlated errors across multiple dimensions. The number of observations used in the regressions by year is as follows: 2016 – 507,184; 2017 – 522,510; 2018 – 542,462; 2019 – 545,300; 2020 – 533,694; 2021 – 544,008; 2022 – 523,018; 2023 – 518,118; and 2024 – 536,110.

distance elasticity of arm's-length exports converges toward that of related-party trade, as arm's-length exports become less distance-elastic over time (Figure 8). This suggests that the evolving pattern is not primarily driven by firm turnover. Instead, it appears to reflect changes in the export behavior of incumbent firms rather than changes in the composition of firms in the sample.

Figure 8: Robustness to Firm Dynamics



Notes: Results from a restricted sample of MNEs that remain actively exporting throughout 2016–2024. Robust standard errors are multi-way clustered by MNE affiliate, source country, product, and destination to account for correlated errors across multiple dimensions.

7 Discussion

7.1 Why might related-party trade be less distance-sensitive?

I show that related-party exports are systematically less sensitive to geographic distance than arm's-length exports until very recent years. This pattern is consistent with several mechanisms emphasized in the existing literature, including lower search and matching frictions, fewer contractual and monitoring problems within firm boundaries, and superior internal coordination and logistics in multinational production networks (Rauch, 1999; Nunn, 2007; Antràs, 2005; Atalay et al., 2019; Bartelme and Ziv, 2024). The evidence in this paper does not separately distinguish among these channels, so this section discusses related explanations in light of the descriptive patterns in the data.

In gravity settings, geographic distance captures a broad set of frictions (Anderson and van Wincoop, 2004), and the observed effect of distance on trade is generally too

large to be explained by freight costs alone ([Head and Mayer, 2014](#)). The discussion below therefore does not attempt to isolate a single dominant channel. Instead, it relates the smaller distance elasticity of related-party trade to two complementary ideas from the literature: first, that firm boundaries help govern transactions that are harder to sustain through markets; and second, that these advantages may be especially strong for parents with broader international networks.

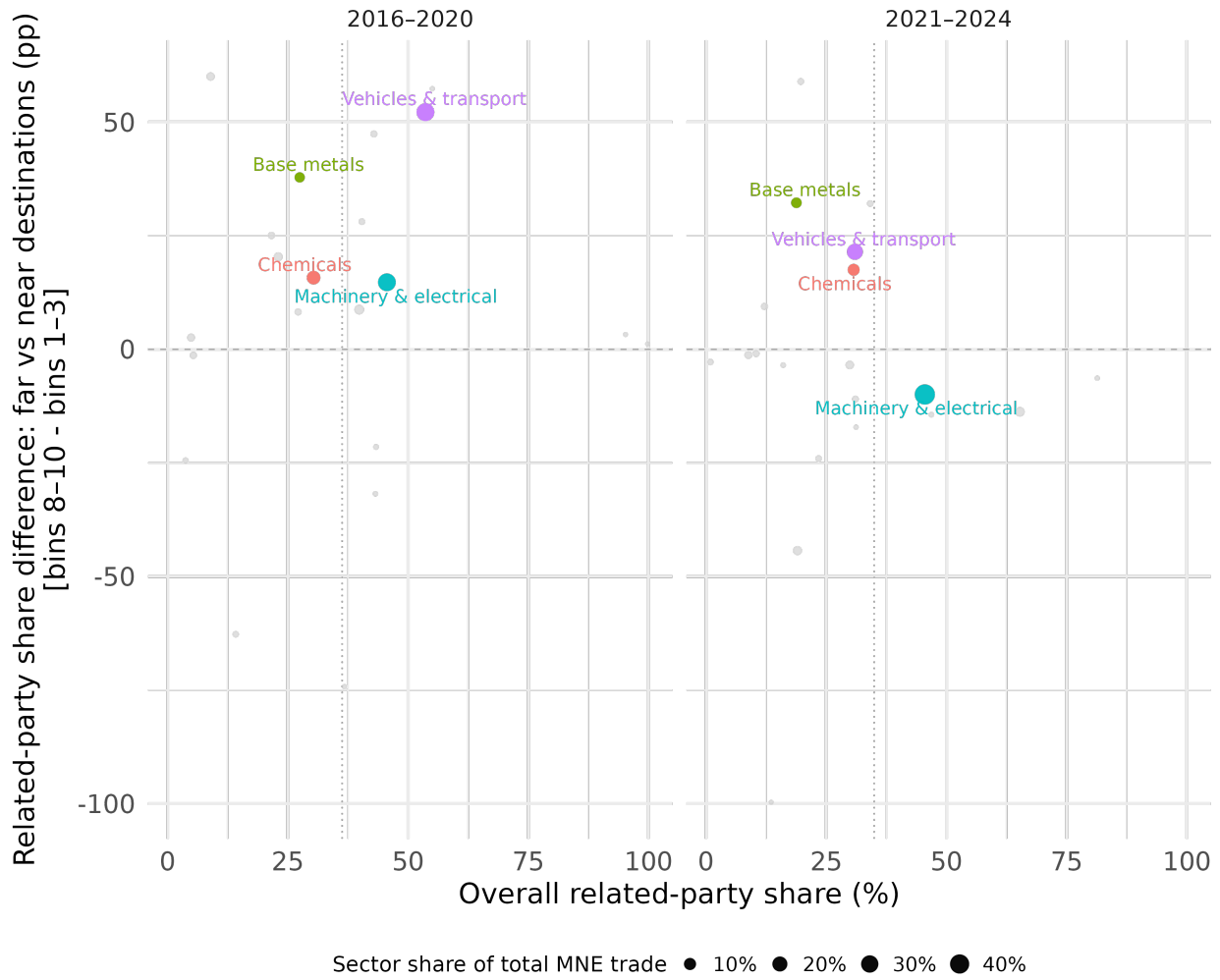
The incomplete-contracts literature views firm boundaries as an allocation of residual control rights when contracts cannot specify all relevant contingencies ex ante ([Aghion and Holden, 2011](#)). In international settings, this logic implies that contractual frictions can limit the fragmentation of production across borders and make within-firm organization relatively more attractive for transactions involving relationship-specific inputs ([Antràs, 2005](#); [Nunn, 2007](#)). This mechanism is relevant here because many related-party shipments involve intermediate inputs that are customized, quality-sensitive, or require ongoing adjustment between the buyer and seller. These transactions are often harder to govern through arm's-length contracts, since important contingencies cannot be fully specified or enforced ex ante, and geographic distance can further worsen contractual frictions. In such cases, within-firm trade can serve as a governance mechanism, allowing firms to manage transactions that are harder to sustain through markets.

This view is also consistent with evidence from [Atalay et al. \(2019\)](#), who show that ownership boundaries matter for the geography of trade. They estimate that domestic within-firm trade costs in the United States are about 40% lower than arm's-length trade costs at a given distance. This result is not directly comparable to mine because their framework embeds a trade-off between distance costs and ownership advantages, whereas my framework assumes a distance-dependent advantage from common ownership and identifies the gap in distance elasticities separately. Nevertheless, the broader interpretation that firm boundaries reduce the effective burden of distance is consistent with my baseline findings. These frictions are often associated with particular types of inputs, especially those involving relationship-specific investments or greater technological complexity. The sector-level descriptive patterns are in line with this explanation. The strongest distance premia²² appear in large manufacturing sectors such as vehicles, base metals, machinery, and chemicals, where production is likely to be input-specific, as shown in [Figure 9](#).

Another related perspective comes from recent work on the internal geography of firms. [Bartelme and Ziv \(2024\)](#) argue that firms face internal costs of geographic expansion, including the transport of intermediates and management or coordination costs that increase with distance. They show that large firms are more geographically dispersed and that plant outcomes decline less with distance within large firms than within small firms. In their model, more productive firms choose ex ante to reduce the distance penalty

²²Here, the distance premium refers to the difference in related-party share between distant and nearby destinations.

Figure 9: Sectoral Differences in the Distance Pattern of Related-Party Trade - India

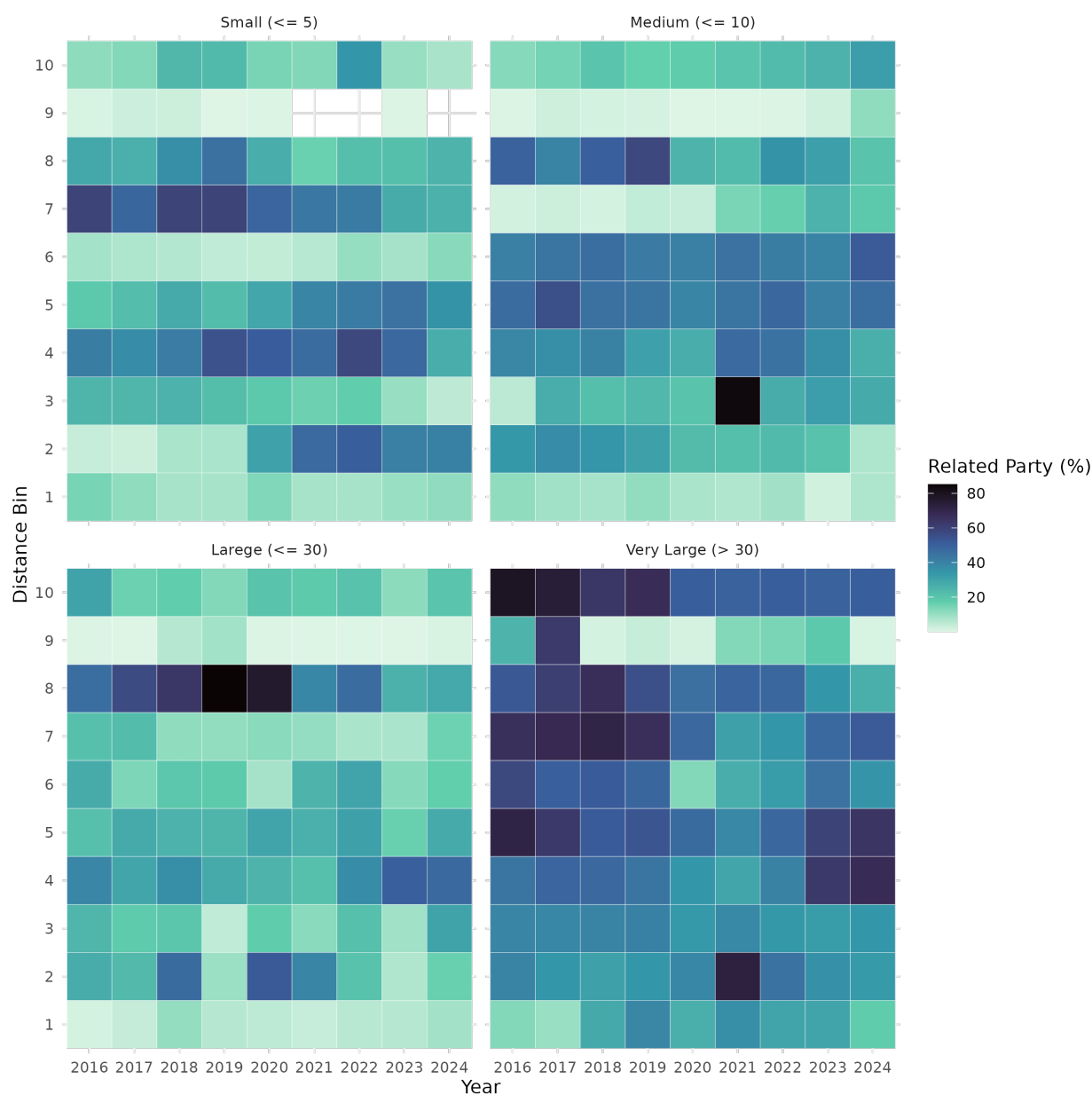


Notes: Each bubble represents an HS Section. The x-axis reports the sector's overall related-party share, defined as the share of exports conducted within firm boundaries relative to total exports. The y-axis reports the difference in related-party share (% points) between distant and nearby destinations, calculated as the related-party share in distance bins 8–10 minus the share in bins 1–3. Bubble size indicates the sector's share in total multinational exports. Colored bubbles correspond to sectors with larger trade shares (larger than 5% in total MNE trade) and are labeled in the figure, while grey bubbles represent the remaining sectors. The horizontal dashed line indicates zero difference between far and near destinations, and the vertical dashed line indicates the average related-party share across sectors within each period.

they face ex post. This framework provides a useful intuition for why related-party trade may be less sensitive to distance, especially for parents with broader affiliate networks: larger firms may be better able to sustain geographically dispersed internal linkages. This is consistent with the descriptive patterns in my data. In [Figure 10](#), where parent size is measured by the number of countries in which the parent has affiliates, the tendency for related-party exports to be disproportionately directed toward distant markets, as shown in Fact 4, becomes more pronounced with parent size.²³

²³In my sample, very large parent firms account for the majority of total trade flows of MNEs ([Figure A11](#)), and time variation is driven primarily by those firms.

Figure 10: Related Party Trade Share by Distance Bin and Year - Foreign MNEs in India by Parent Firm Size



Notes: This figure illustrates the related-party trade share by distance of MNEs headquartered outside India. The ratio is calculated within each distance bin using aggregated trade values. Parent firm size is measured by the number of countries in which the parent has affiliates. For example, the “Very Large” group includes parent firms with affiliates in more than 30 countries. For India, the seventh distance bin includes West and Southern African countries such as South Africa and Nigeria. The eighth bin covers Canada and the United States, Oceania, and some West African countries.

7.2 Potential Mechanism of Narrowing Gap

To assess whether a search-cost mechanism helps explain the post-2021 decline in the distance-elasticity gap and the concurrent increase in arm's-length trade, I construct buyer-level outcomes for each MNE (parent) $\times$ HS6 product $\times$ destination $\times$ year. If the narrowing gap reflects improved search conditions for distant arm's-length transactions, then distant destinations should exhibit lower buyer concentration and a broader buyer base in the post-2021 period. At the same time, these buyer-based measures should be interpreted with caution because importer identifiers and names may change or be recorded inconsistently over time, even when they refer to the same underlying firm.

For each parent p , product k , destination j , and year t , let x_{bpkjt}^{AL} denote arm's-length sales to buyer b . Total arm's-length sales are given by $X_{pkjt}^{AL} = \sum_b x_{bpkjt}^{AL}$ and buyer shares are $s_{bpkjt} = \frac{x_{bpkjt}^{AL}}{X_{pkjt}^{AL}}$.

I measure buyer concentration using the Herfindahl index,

$$HHI_{pkjt}^{AL} = \sum_b s_{bpkjt}^2, \quad (7.1)$$

and average buyer size,

$$AvgBuyerSize_{pkjt}^{AL} = \frac{X_{pkjt}^{AL}}{N_{pkjt}^{AL}}, \quad (7.2)$$

where N_{pkjt}^{AL} is the number of distinct arm's-length buyers. Because these measures rely on transaction-level importer identifiers, I restrict the buyer-side analysis to India, where buyer coverage is more reliable.

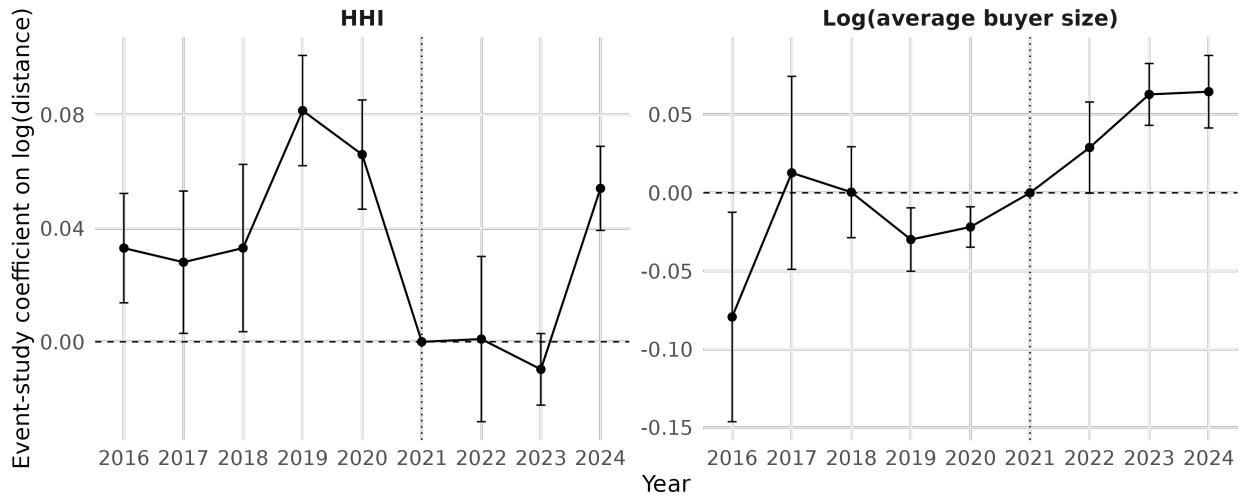
To examine how the relationship between distance and buyer-side outcomes evolves over time, I use MNEs observed in all nine years from 2016 to 2024 and estimate the following event-study specification:

$$Y_{pkjt} = \beta_d \log(dist_j) + \sum_{\tau \neq 2021} \beta_\tau [\log(dist_j) \times \mathbf{1}(t = \tau)] + \Gamma' Z_{jt} + \alpha_{pk} + \alpha_t + \varepsilon_{pkjt}, \quad (7.3)$$

where Y_{pkjt} is one of HHI_{pkjt}^{AL} or $\log(AvgBuyerSize_{pkjt}^{AL})$, $dist_j$ denotes bilateral distance, and Z_{jt} includes destination-year controls such as $\log(gdp_{jt})$, $\log(pop_{jt})$, and fta_{jt} . The fixed effects α_{pk} and α_t denote parent-product and year fixed effects. I omit 2021 as the reference year.

The results in [Figure 11](#) point to a post-2021 change in the buyer-side pattern of distant arm's-length trade. First, the association between distance and buyer concentration weakens after 2021, although the trend is not fully monotonic because the pattern shifts again in 2024. This suggests that, before 2021, exporters were more dependent on a narrow

Figure 11: Evolution of the Distance Slope Relative to 2021 - India



Notes: The figure reports point estimates of β_τ and 95% confidence intervals. All coefficients are estimated by OLS, with standard errors clustered by parent, product, destination, and year.

set of buyers in more distant destinations. Second, the relationship between distance and average buyer size becomes more positive after 2021. In addition, when I decompose changes in arm's-length trade across distance, the increase is concentrated primarily in transactions involving old buyers and old products (see [Figure A12](#)). Taken together, this second pattern is more consistent with the expansion of existing relationships, or possibly improved matching within established buyer networks, than with a decline in initial search costs.

To further evaluate the search-cost channel, I also examine whether the number of new seller-buyer matches increased disproportionately in more distant markets after 2021. I do not find economically or statistically meaningful evidence of such a pattern. Overall, the evidence is insufficient to conclude that improved search conditions are the primary mechanism behind the post-2021 narrowing in the distance-elasticity gap.

8 Conclusion

Using shipment-level Bill of Lading data from India, this paper documents four stylized facts about the export behavior of foreign MNE affiliates. Foreign affiliates export through both related-party and arm's-length channels, and they typically use the two modes for different sets of products. The probability of engaging in related-party trade rises with destination distance, and through 2020 related-party exports were disproportionately concentrated in distant markets. This pattern, however, has weakened in recent years. Taken together, these facts point to systematic differences in how distance shapes within-firm and arm's-length trade, and suggest that this difference has narrowed over time.

Motivated by these patterns, I derive a firm-level gravity equation with mode-specific distance elasticities. Estimating this equation using Poisson pseudo-maximum likelihood, I find that related-party exports are less sensitive to distance than arm's-length exports. This within-firm distance advantage, however, has weakened in recent years.

These results imply that internal firm trade remains an important margin through which multinational firms manage distance-related trade frictions. At the same time, the recent weakening of this advantage suggests that the relative benefits of internal trade have declined, consistent with a broader shift toward greater reliance on external market relationships in global production.

This paper also has several limitations. Most importantly, it does not pin down the mechanism behind the recent narrowing in the distance-elasticity gap between related-party and arm's-length trade. While the finding that related-party exports are less sensitive to distance is consistent with the existing literature and supported by the results in this paper, the post-2021 convergence may reflect changes in search and matching frictions, improvements in the contractibility, or broader changes in multinational production networks. The current analysis cannot distinguish among these explanations.

This paper also has several limitations. Most importantly, it does not pin down the mechanism behind the recent narrowing in the distance-elasticity gap between related-party and arm's-length trade. I examine buyer-seller relationship patterns to shed light on the search and matching channel, but the results do not point to a clear explanation for the post-2021 convergence. In addition, this exercise is limited by the fact that importer identifiers and names may change over time even when they refer to the same underlying firm. A more careful reconciliation of importer identities would therefore be needed to trace buyer relationships reliably. Another potential avenue is product-level heterogeneity in contractual frictions, using measures such as Rauch's classification of differentiated goods (Rauch, 1999) or Nunn's contract-intensity index (Nunn, 2007). If firm boundaries are especially preferred when market transactions are harder to govern, then the decline

in the within-firm distance advantage should be more pronounced for goods for which those governance advantages were previously more important.

Establishing this mechanism more directly to explain the pattern in the data will require both a richer theoretical framework and sharper empirical identification. On the theory side, an important next step is to extend the framework so that firms can adjust their organizational choices in response to changes in trade costs. On the empirical side, future work should pursue identification strategies that can more directly isolate the sources of the post-2021 narrowing in the distance-elasticity gap.

Lastly, a separate pattern in the data is the sharp recent increase in exports by home-country MNEs, both within and across firm boundaries. This may reflect broader supply-chain reorganization or changes in demand conditions surrounding India. Understanding the forces behind that expansion is also an important direction for future research.

References

- Aghion, Philippe and Richard Holden**, “Incomplete Contracts and the Theory of the Firm: What Have We Learned over the Past 25 Years?,” *Journal of Economic Perspectives*, June 2011, 25 (2), 181–197.
- Alfaro, Laura and Maggie X. Chen**, “Selection and Market Reallocation: Productivity Gains from Multinational Production,” *American Economic Journal: Economic Policy*, May 2018, 10 (2), 1–38.
- , **Paola Conconi, Fariha Kamal, and Zachary Kroff**, “Trade within Multinational Boundaries,” *National Bureau of Economic Research*, June 2025.
- Alfaro-Ureña, Alonso, Isabela Manelici, and Jose P Vasquez**, “The Effects of Joining Multinational Supply Chains: New Evidence from Firm-to-Firm Linkages,” *The Quarterly Journal of Economics*, July 2022, 137 (3), 1495–1552.
- Alfaro-Urena, Alonso, Isabela Manelici, and Jose P. Vasquez**, “The Effects of Multinationals on Workers: Evidence from Costa Rican Microdata,” *Working Papers*, April 2021, (285).
- Alviarez, Vanessa and Ayhab F. Saad**, “Multinational Production and Intra-Firm Trade,” *Review of International Economics*, 2023, 31 (3), 826–853.
- Anderson, James E. and Eric van Wincoop**, “Trade Costs,” *Journal of Economic Literature*, September 2004, 42 (3), 691–751.
- Antràs, Pol**, “Incomplete Contracts and the Product Cycle,” *American Economic Review*, September 2005, 95 (4), 1054–1073.
- and **Stephen R. Yeaple**, “Chapter 2 - Multinational Firms and the Structure of International Trade,” in Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, eds., *Handbook of International Economics*, Vol. 4 of *Handbook of International Economics*, Elsevier, 2014, pp. 55–130.
- , **Evgenii Fadeev, Teresa C. Fort, and Felix Tintelnot**, “Exporting, Global Sourcing, and Multinational Activity: Theory and Evidence from the United States,” *The Review of Economics and Statistics*, April 2024, pp. 1–48.
- Atalay, Enghin, Ali Hortaçsu, Mary Jialin Li, and Chad Syverson**, “How Wide Is the Firm Border?,” *The Quarterly Journal of Economics*, November 2019, 134 (4), 1845–1882.
- Balsvik, Ragnhild**, “Is Labor Mobility a Channel for Spillovers from Multinationals? Evidence from Norwegian Manufacturing,” *The Review of Economics and Statistics*, 2011, 93 (1), 285–297.
- Bartelme, Dominick and Oren Ziv**, “The Internal Geography of Firms,” *Journal of International Economics*, March 2024, 148, 103889.
- Bas, Maria, Thierry Mayer, and Mathias Thoenig**, “From Micro to Macro: Demand, Supply, and Heterogeneity in the Trade Elasticity,” *Journal of International Economics*, September 2017, 108, 1–19.

- Berthelon, Matias and Caroline Freund**, “On the Conservation of Distance in International Trade,” *Journal of International Economics*, July 2008, 75 (2), 310–320.
- Bombarda, Pamela and Stefania Marcassa**, “Welfare and Trade Elasticity with Multinational Production,” *The World Economy*, 2020, 43 (2), 388–411.
- Bruno, Valentina and Hyun Song Shin**, “Dollar and Exports,” *The Review of Financial Studies*, August 2023, 36 (8), 2963–2996.
- Cadestin, Charles, Koen De Backer, Isabelle Desnoyers-James, Sébastien Miroudot, Ming Ye, and Davide Rigo**, “Multinational Enterprises and Global Value Chains: New Insights on the Trade-Investment Nexus,” OECD Science, Technology and Industry Working Papers 2018/05, OECD March 2018.
- Chen, Kaicheng, Robert S. Martin, and Jeffrey M. Wooldridge**, “Another Look at the Linear Probability Model and Nonlinear Index Models,” October 2023.
- Conconi, Paola, Fabrizio Leone, Glenn Magerman, and Catherine Thomas**, “Multinational Networks and Trade Participation,” *Working Paper*, 2025, (456).
- Davies, Ronald B., Julien Martin, Mathieu Parenti, and Farid Toubal**, “Knocking on Tax Haven’s Door: Multinational Firms and Transfer Pricing,” *The Review of Economics and Statistics*, March 2018, 100 (1), 120–134.
- Delpeuch, Samuel, Ana Cinta González Cabral, Felix Hugger, and Pierce O’Reilly**, “MNE Business Functions and Corporate Taxation,” OECD Taxation Working Papers January 2025.
- Dixit, Avinash K. and Joseph E. Stiglitz**, “Monopolistic Competition and Optimum Product Diversity,” *The American Economic Review*, 1977, 67 (3), 297–308.
- Do, Anh, Sharat Ganapati, Woan Foong Wong, and Oren Ziv**, “Transshipment Hubs, Trade, and Supply Chains,” *Working Paper*, 2025.
- Eaton, Jonathan and Samuel Kortum**, “Technology, Geography, and Trade,” *Econometrica*, 2002, 70 (5), 1741–1779.
- Finger, J. M. and M. E. Kreinin**, “A Measure of ‘Export Similarity’ and Its Possible Uses,” *The Economic Journal*, December 1979, 89 (356), 905–912.
- Flaen, Aaron, Flora Haberkorn, Logan Lewis, Anderson Monken, Justin Pierce, Rosemary Rhodes, and Madeleine Yi**, “Bill of Lading Data in International Trade Research with an Application to the COVID-19 Pandemic,” *Review of International Economics*, 2023, 31 (3), 1146–1172.
- Ganapati, Sharat, Woan Foong Wong, and Oren Ziv**, “Entrepôt: Hubs, Scale, and Trade Costs,” *American Economic Journal: Macroeconomics*, October 2024, 16 (4), 239–278.
- Garetto, Stefania, Xiao Ma, Lindsay Oldenski, and Natalia Ramondo**, “Multinational Expansion in Time and Space,” *Working Paper*, 2024.
- Garg, Shresth, Pulak Ghosh, and Brandon Joel Tan**, “Within Firm Supply Chains: Evidence from India,” *Journal of International Economics*, September 2023, 144, 103793.

- Ge, Ying, Huiwen Lai, and Susan Chun Zhu**, “Multinational Price Premium,” *Journal of Development Economics*, July 2015, 115, 181–199.
- Gourieroux, C., A. Monfort, and A. Trognon**, “Pseudo Maximum Likelihood Methods: Applications to Poisson Models,” *Econometrica*, 1984, 52 (3), 701–720.
- , —, and —, “Pseudo Maximum Likelihood Methods: Theory,” *Econometrica*, 1984, 52 (3), 681–700.
- Head, Keith and Thierry Mayer**, “Gravity Equations: Workhorse, Toolkit, and Cookbook,” in “Handbook of International Economics,” Vol. 4, Elsevier, 2014, pp. 131–195.
- and —, “Brands in Motion: How Frictions Shape Multinational Production,” *American Economic Review*, September 2019, 109 (9), 3073–3124.
- Hong, YongKi**, “Related-Party Trades in Vertical Integration,” *Working Paper*, 2023.
- Irrarrazabal, Alfonso, Andreas Moxnes, and Luca David Opromolla**, “The Margins of Multinational Production and the Role of Intrafirm Trade,” *Journal of Political Economy*, February 2013, 121 (1), 74–126.
- Javorcik, Beata Smarzyska**, “Does Foreign Direct Investment Increase the Productivity of Domestic Firms? In Search of Spillovers Through Backward Linkages,” *American Economic Review*, June 2004, 94 (3), 605–627.
- Keller, Wolfgang**, “Chapter 19 - International Trade, Foreign Direct Investment, and Technology Spillovers,” in Bronwyn H. Hall and Nathan Rosenberg, eds., *Handbook of the Economics of Innovation*, Vol. 2 of *Handbook of the Economics of Innovation, Volume 2*, North-Holland, January 2010, pp. 793–829.
- Krugman, Paul**, “Scale Economies, Product Differentiation, and the Pattern of Trade,” *The American Economic Review*, 1980, 70 (5), 950–959.
- Leino, Topias and Maja Gavrilovic**, “Foreign Direct Investment Increased to a Record \$41 Trillion,” IMF Blog February 2025.
- Lipsey, Robert E and Fredrik Sjöholm**, “Foreign Direct Investment, Education and Wages in Indonesian Manufacturing,” *Journal of Development Economics*, February 2004, 73 (1), 415–422.
- Moran, Theodore H.**, “Foreign Direct Investment and Development: Launching a Second Generation of Policy Research: Avoiding the Mistakes of the First, Reevaluating Policies for Developed and Developing Countries,” *Peterson Institute Press: All Books*, 2011.
- , **Edward M. Graham, and Magnus Blomstrom**, “Does Foreign Direct Investment Promote Development?,” *Peterson Institute Press: All Books*, 2005.
- Nunn, Nathan**, “Relationship-Specificity, Incomplete Contracts, and the Pattern of Trade*,” *The Quarterly Journal of Economics*, May 2007, 122 (2), 569–600.
- Ramondo, Natalia, Veronica Rappoport, and Kim J. Ruhl**, “Intrafirm Trade and Vertical Fragmentation in U.S. Multinational Corporations,” *Journal of International Economics*, January 2016, 98, 51–59.

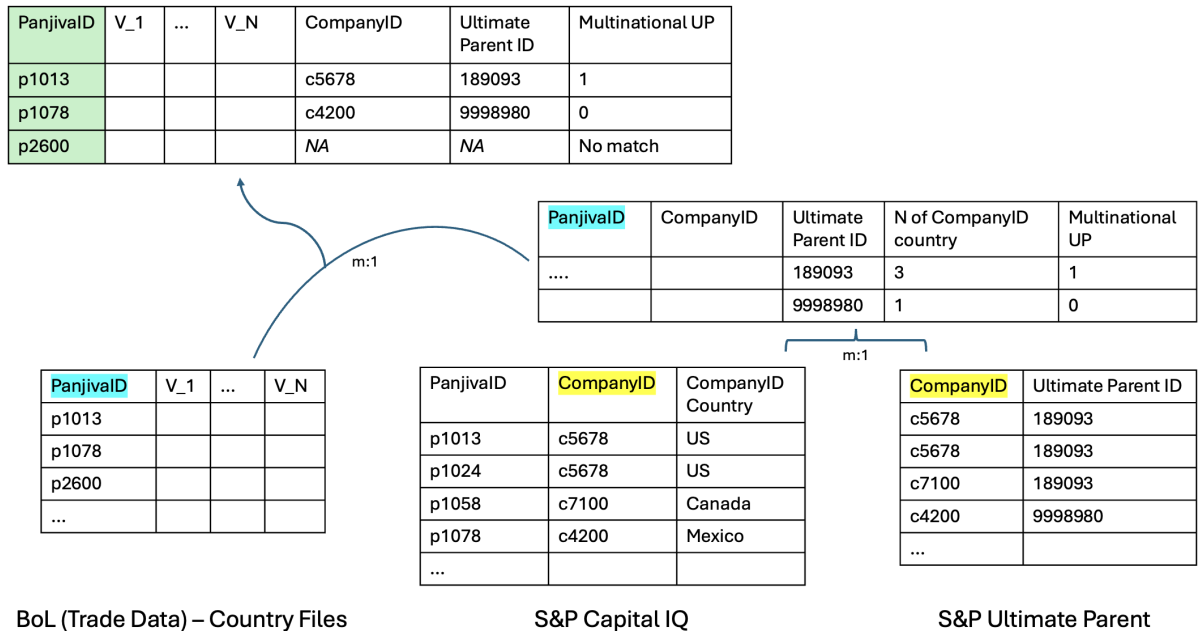
- Rauch, James E.**, “Networks versus Markets in International Trade,” *Journal of International Economics*, June 1999, 48 (1), 7–35.
- Ruhl, Kim J.**, “How Well Is US Intrafirm Trade Measured?,” *American Economic Review*, May 2015, 105 (5), 524–529.
- Setzler, Bradley and Felix Tintelnot**, “The Effects of Foreign Multinationals on Workers and Firms in the United States*,” *The Quarterly Journal of Economics*, August 2021, 136 (3), 1943–1991.
- Silva, J. M. C. Santos and Silvana Tenreyro**, “The Log of Gravity,” *The Review of Economics and Statistics*, November 2006, 88 (4), 641–658.
- and —, “Further Simulation Evidence on the Performance of the Poisson Pseudo-Maximum Likelihood Estimator,” *Economics Letters*, August 2011, 112 (2), 220–222.
- S&P Global**, “Panjiva Supply Chain Intelligence User Guide,” Technical Report 2024.
- UNCTAD**, *World Investment Report 2024: Investment Facilitation and Digital Government*, Geneva: United Nations Conference on Trade and Development, 2024.
- Wang, Zi**, “Headquarters Gravity: How Multinationals Shape International Trade,” *Journal of International Economics*, July 2021, 131, 103477.
- Wooldridge, Jeffrey M.**, “Quasi-Likelihood Methods for Count Data,” in “Handbook of Applied Econometrics Volume 2: Microeconomics,” John Wiley & Sons, Ltd, 1999, chapter 8, pp. 321–368.
- Wooldridge, Jeffrey M.**, “Simple Approaches to Nonlinear Difference-in-Differences with Panel Data,” *The Econometrics Journal*, September 2023, 26 (3), C31–C66.

Appendix

A Data Details

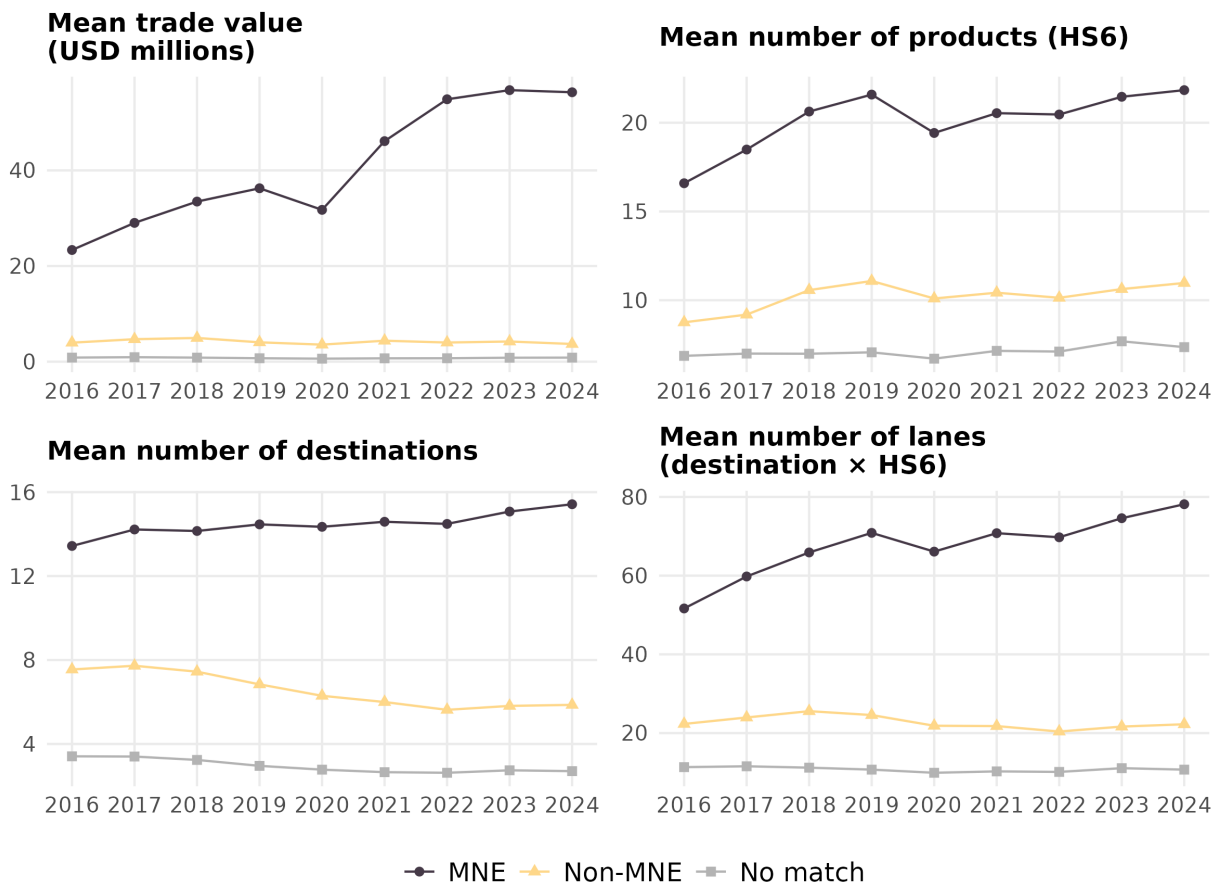
A.1 Process of Identifying Multinational

Figure A1: Data Manipulation and Database Merge Process



Notes:

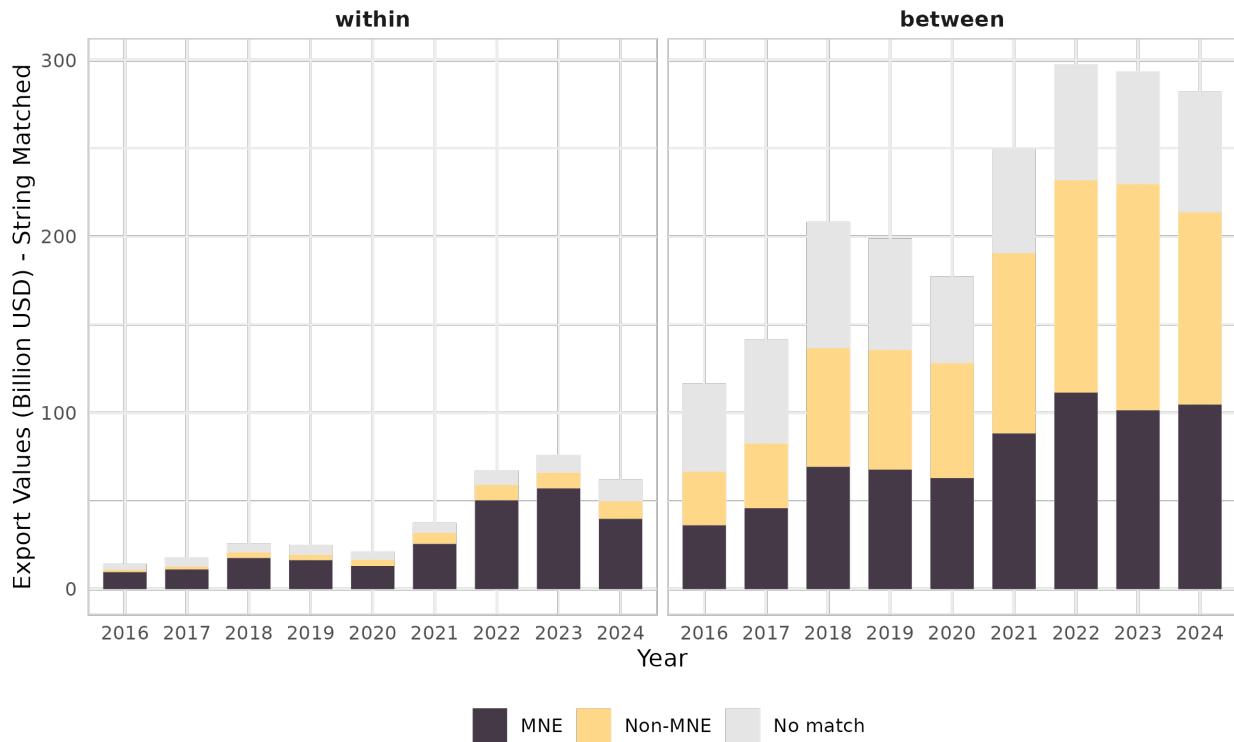
Figure A2: Average Shipper Characteristics by MNE Status



Notes: Each panel shows the average outcome per shipper by year and MNE classification group. For each year, I first construct shipper-year level measures, including trade value, number of HS6 products, number of destination markets, and number of destination-product lanes, and then average these measures across shippers within each MNE classification group.

A.2 Related Party and Arm's length Export Matching

Figure A3: Related Party and Arm's length Export Matching



Notes: As explained in the main text, MNE status is determined by the parent firm's classification. "No match" refers to firms that are either not found in the S&P firm database or lack parent-firm information. Related-party trade is identified through string matching between exporter and importer names. Transactions with missing exporter or importer firm names are also removed. As a result, the total export value reported in this figure (the sum of within and between) is lower than that in [Figure 1](#). In the labels, "within" denotes related-party exports and "between" denotes arm's-length exports.

A.3 Harmonized System (HS) 2 digit codes and Section group

This subsection provides further details on MNEs' exports by product group. Product groups are classified according to the Harmonized System (HS) sections, as defined by the World Customs Organization (WCO). The last three sections in the table (Table A1), covering arms, miscellaneous manufactured goods, works of art, and unclassified items, are excluded.

Table A1: HS 2-digit Codes and Corresponding Sections

HS 2-digit Codes	Section Description
01–05	Live Animals & Animal Products
06–14	Vegetable Products
15	Animal or Vegetable Fats and Oils
16–24	Prepared Foodstuffs; Beverages, Spirits & Tobacco
25–27	Mineral Products
28–38	Products of the Chemical or Allied Industries
39–40	Plastics & Articles Thereof; Rubber & Articles Thereof
41–43	Raw Hides, Skins, Leather, Furskins & Articles Thereof
44–46	Wood & Articles of Wood; Cork & Articles of Cork
47–49	Pulp of Wood or Other Fibrous Material; Paper & Paperboard
50–63	Textiles & Textile Articles
64–67	Footwear, Headgear, Umbrellas & Related Articles
68–70	Articles of Stone, Plaster, Cement, Ceramic & Glass
71	Natural or Cultured Pearls, Precious Stones & Metals
72–83	Base Metals & Articles Thereof
84–85	Machinery & Mechanical Appliances; Electrical Equipment
86–89	Vehicles, Aircraft, Vessels & Transport Equipment
90–92	Optical, Measuring, Medical Instruments
93	Arms & Ammunition; Parts & Accessories
94–96	Miscellaneous Manufactured Articles
97	Works of Art, Collectors' Pieces & Antiques

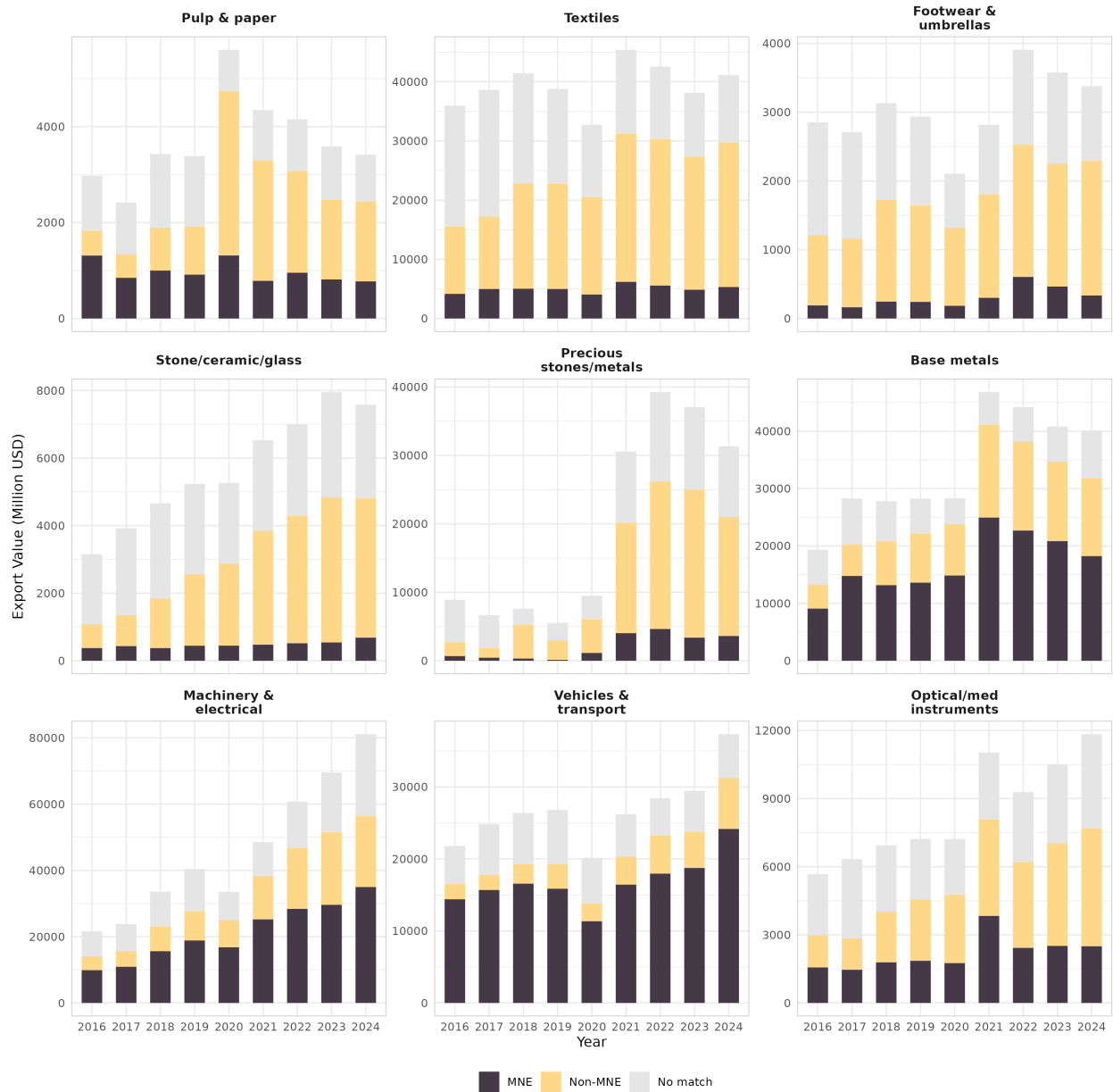
Note: HS 2-digit codes are grouped according to the Harmonized System (HS) Sections, as defined by the World Customs Organization (WCO). Section descriptions have been slightly abbreviated for clarity.

Figure A4: Export from India by Firm Type and By Product Group



Notes: Y-axis is export value in million USD. Each product group is based on the Harmonized System (HS) Sections, as defined by the World Customs Organization (WCO). Details on the group classification can be found in [Table A1](#).

Figure A5: Export from India by Firm Type and By Product Group (Continued)



Notes: Y-axis is export value in million USD. Each product group is based on the Harmonized System (HS) Sections, as defined by the World Customs Organization (WCO). Details on the group classification can be found in [Table A1](#).

A.4 Process of Identifying Related Party Trade

In this section, I outline how I match shipper and consignee names to determine whether a shipment is related-party trade.

Entity cleaning. I harmonize shipper (exporter, sender) and consignee (importer, receiver) names before matching. Using case-insensitive regex, I remove country-specific legal forms and common suffixes, then strip trailing digits, stray punctuation and whitespace, and orphan trailing letters. Examples of removed tokens include *S.A.S.*, *S.A.C.*, *C.V.*, *S.A.*, *R.L.*, *DE*, as well as *LTD/Ltda*, *LLC/Llc*, *Inc/INC*, *Pvt./PVT*, *Co./Corporation*, *International/Internacional*, *Trading*.²⁴

Observations with empty cleaned names are dropped, and I use the cleaned names to compute similarity between two firms as follows:

Similarity metrics. For each shipment, I compute two complementary similarity measures:

- **Jaro–Winkler similarity** ($JW \in [0, 1]$), capturing character-level closeness between full cleaned strings (implemented with parameter $p = 0.1$).
- **Jaccard token similarity** ($J \in [0, 1]$), computed on sets of word tokens extracted from the cleaned strings.
 - For each pair of cleaned names a and b , I lowercase both strings and tokenize on non-word boundaries, remove empty tokens, and then form sets A and B of unique tokens. The Jaccard index is defined as

$$J(a, b) = \frac{|A \cap B|}{|A \cup B|},$$

This token-based Jaccard index is robust to reordered or duplicated words (e.g., “Cute Orange” vs. “Orange Cute”). I compute this index for all shipment pairs.²⁵

Decision rule. A shipment is labeled **within-firm** (“related-party”) if any of the following conditions is satisfied:

$$\begin{aligned} JW &\geq 0.80, \quad \text{or} \\ 0.70 &\leq JW < 0.80 \text{ and } J > 0.10, \quad \text{or} \\ J &\geq 0.50. \end{aligned}$$

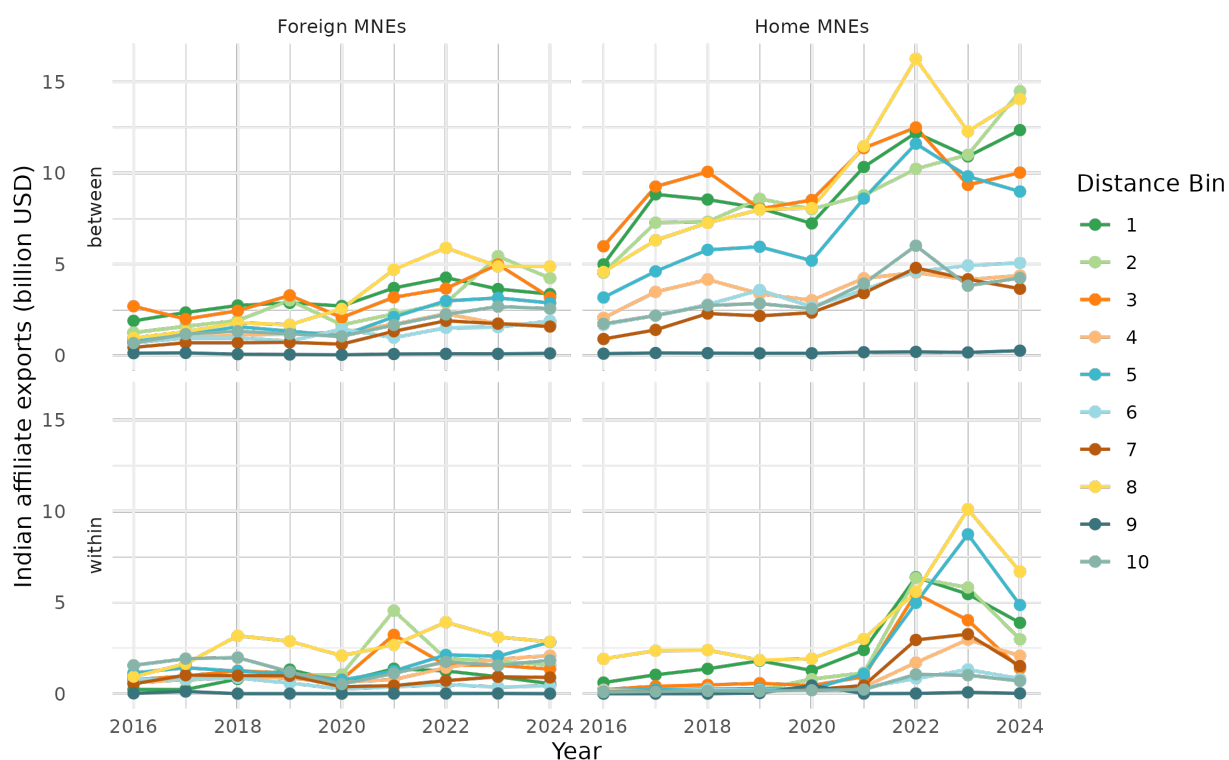
²⁴Token lists differ by country (Mexico vs. India) to reflect local conventions.

²⁵The baseline uses an unweighted set Jaccard for transparency and speed.

If none of the conditions hold, the shipment is labeled **between-firm** (“Arm’s Length”, or “across-firm”).

Discussion. The rule first prioritizes high character-level similarity (JW), then allows medium JW when modest token overlap exists ($J > 0.10$), and finally captures cases with strong token overlap ($J \geq 0.50$) even when character-level similarity is lower, such as reordered, abbreviated, or partially truncated firm names. Because Jaro–Winkler and Jaccard capture different linguistic dimensions— JW measures fine-grained spelling similarity, while J focuses on shared word content—they complement each other: the former performs better for short or highly similar strings, whereas the latter is more robust to reordering and structural variation. Combining these metrics therefore allows robust detection of within-firm shipments across common spelling variants, token reorderings, and language differences. Thresholds (0.80, 0.70, 0.10, 0.50) were chosen conservatively through manual inspection and iterative trial and error to balance precision (avoiding false positives) and recall (capturing genuine within-firm matches). Future work may explore hybrid or machine-learning approaches, including natural language-based name matching. However, developing such models requires a sufficiently comprehensive and accurately labeled set of multinational firms and their subsidiaries—data that are only partially available through existing commercial sources such as Orbis or Panjiva. Building and validating such a dataset remains an important avenue for future research.

Figure A6: Related Party Trade Volume by Distance Bin - India



Notes: The x-axis indicates years and the y-axis reports the value of Indian affiliate exports in billions of U.S. dollars. Each colored line corresponds to a destination distance bin, where bins 1–10 group destinations from the shortest to the longest bilateral distance from India. Panels separate exports by foreign and home MNEs and by transaction mode. The legend indicates the distance bin associated with each line.

B Details on Theoretical Framework

B.1 Demand and Price Index (Multi-Sector)

Final-good producers in j aggregate *sectors* with a Cobb–Douglas structure and, *within each sector* g , aggregate a continuum of varieties $\omega \in \Omega^g$ with CES elasticity $\sigma_g > 1$.

$$Q_j^g = \left(\int_{\Omega^g} q_j^g(\omega)^{\frac{\sigma_g-1}{\sigma_g}} d\omega \right)^{\frac{\sigma_g}{\sigma_g-1}}, \quad Q_j = \prod_{g \in \mathcal{G}} (Q_j^g)^{\alpha_g}, \quad \alpha_g > 0, \sum_g \alpha_g = 1. \quad (\text{B.1})$$

Given sector- g prices $p_j^g(\omega)$, the producer chooses $\{q_j^g(\omega)\}_{\omega \in \Omega^g}$ to minimize cost for a target level of Q_j^g :

$$\min_{\{q_j^g(\cdot) \geq 0\}} \int_{\Omega^g} p_j^g(\omega) q_j^g(\omega) d\omega \quad \text{s.t.} \quad \left(\int_{\Omega^g} q_j^g(\omega)^{\frac{\sigma_g-1}{\sigma_g}} d\omega \right)^{\frac{\sigma_g}{\sigma_g-1}} \geq Q_j^g. \quad (\text{B.2})$$

Solving the sector- g buyer's problem. Let λ_j^g denote the multiplier on the sector- g output constraint. The Lagrangian is

$$\mathcal{L}^g = \int_{\Omega^g} p_j^g(\omega) q_j^g(\omega) d\omega + \lambda_j^g \left[Q_j^g - \left(\int_{\Omega^g} q_j^g(\omega)^{\frac{\sigma_g-1}{\sigma_g}} d\omega \right)^{\frac{\sigma_g}{\sigma_g-1}} \right].$$

The first-order condition for any ω with $q_j^g(\omega) > 0$ is

$$p_j^g(\omega) = \lambda_j^g \left(\int_{\Omega^g} q_j^g(\tilde{\omega})^{\frac{\sigma_g-1}{\sigma_g}} d\tilde{\omega} \right)^{\frac{1}{\sigma_g-1}} \cdot q_j^g(\omega)^{-\frac{1}{\sigma_g}}.$$

From the CES definition in (B.1),

$$\int_{\Omega^g} q_j^g(\omega)^{\frac{\sigma_g-1}{\sigma_g}} d\omega = (Q_j^g)^{\frac{\sigma_g-1}{\sigma_g}}.$$

Substituting into the FOC gives

$$p_j^g(\omega) = \lambda_j^g (Q_j^g)^{\frac{1}{\sigma_g}} q_j^g(\omega)^{-\frac{1}{\sigma_g}}.$$

Rearranging yields

$$q_j^g(\omega) = (\lambda_j^g)^{\sigma_g} Q_j^g p_j^g(\omega)^{-\sigma_g}. \quad (\text{B.3})$$

Plugging (B.3) into the aggregation constraint gives:

$$(Q_j^g)^{\frac{\sigma_g-1}{\sigma_g}} = (\lambda_j^g)^{\sigma_g-1} (Q_j^g)^{\frac{\sigma_g-1}{\sigma_g}} \int_{\Omega^g} p_j^g(\omega)^{1-\sigma_g} d\omega,$$

so that

$$(\lambda_j^g)^{\sigma_g-1} = \left(\int_{\Omega^g} p_j^g(\omega)^{1-\sigma_g} d\omega \right)^{-1}.$$

We can express $\lambda_j^g = (P_j^g)^{-1}$ by introducing the sectoral Dixit–Stiglitz price index²⁶: substituting $\lambda_j^g = (P_j^g)^{-1}$ back into (B.3) gives the sectoral Hicksian demand,

$$q_j^g(\omega) = Q_j^g \left(\frac{p_j^g(\omega)}{P_j^g} \right)^{-\sigma_g}. \quad (\text{B.4})$$

Minimum cost and expenditure (sectoral and aggregate). Plugging (B.4) into sectoral total cost gives

$$C_j^g(Q_j^g) = \int_{\Omega^g} p_j^g(\omega) q_j^g(\omega) d\omega = Q_j^g (P_j^g)^{\sigma_g} \int_{\Omega^g} p_j^g(\omega)^{1-\sigma_g} d\omega = Q_j^g P_j^g.$$

Thus P_j^g is the unit expenditure function in sector g . With Cobb–Douglas across sectors,

$$P_j = \prod_{g \in \mathcal{G}} (P_j^g)^{\alpha_g}, \quad E_j^g = \alpha_g E_j,$$

and the sectoral Marshallian demand is

$$q_j^g(\omega) = \frac{E_j^g}{P_j^g} \left(\frac{p_j^g(\omega)}{P_j^g} \right)^{-\sigma_g}.$$

B.2 Price Index Derivation (Sector Level)

This derivation follows [Eaton and Kortum \(2002\)](#) within each sector g . Suppliers are indexed by (n, i) with $n \in N$ and $i \in S_n$, and modes are indexed by $m \in \{W, B\}$. The

²⁶The CES price index

$$P_j^g = \left(\int_{\Omega^g} p_j^g(\omega)^{1-\sigma_g} d\omega \right)^{\frac{1}{1-\sigma_g}}$$

is the unit expenditure function associated with the CES aggregator, representing the minimum cost of assembling one unit of the composite good from a continuum of varieties ([Dixit and Stiglitz, 1977](#)). This formulation is widely adopted in both monopolistic-competition and Ricardian trade models and seminal examples include [Krugman \(1980\)](#) and [Eaton and Kortum \(2002\)](#). Intuitively, the Lagrange multiplier λ_j in the cost-minimization problem equals the inverse of the unit price, $\lambda_j^g = 1/P_j^g$, as it represents the marginal cost of producing one additional unit of the CES composite.

delivered price in sector g is

$$p_{nij}^{g,m}(\omega) = \frac{c_n^g}{z_{ni}^m(\omega)} \tau_{nj}^{g,m},$$

where $c_n^g > 0$ and $\tau_{nj}^{g,m} > 0$ are sector- and mode-specific. Productivity draws are i.i.d. Fréchet and (by assumption) common across g :

$$\Pr(z_{ni}^m(\omega) \leq z) = \exp(-T_{ni} z^{-\theta}), \quad T_{ni} > 0, \theta > 1.$$

Define

$$\kappa_{nij}^{g,m} \equiv T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta}, \quad \Phi_j^g \equiv \sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} \kappa_{nij}^{g,m}.$$

In equilibrium, the sectoral price index can be rewritten as:

$$(P_j^g)^{1-\sigma_g} = \int_0^\infty p^{1-\sigma_g} dG_j^g(p) = \theta \Phi_j^g \int_0^\infty p^{\theta-\sigma_g} \exp\{-\Phi_j^g p^\theta\} dp,$$

where $G_j^g(p)$ is the cdf of the sector- g minimum price (derived below). Letting $x \equiv p^\theta \Phi_j^g$ and $dx = \theta \Phi_j^g p^{\theta-1} dp$, we obtain

$$(P_j^g)^{1-\sigma_g} = (\Phi_j^g)^{\frac{1-\sigma_g}{\theta}} \int_0^\infty x^{\frac{1-\sigma_g}{\theta}} e^{-x} dx = (\Phi_j^g)^{\frac{1-\sigma_g}{\theta}} \Gamma\left(\frac{\theta+1-\sigma_g}{\theta}\right).$$

Therefore,

$$P_j^g = (\Phi_j^g)^{-\frac{1}{\theta}} \left[\Gamma\left(\frac{\theta+1-\sigma_g}{\theta}\right) \right]^{\frac{1}{1-\sigma_g}}, \quad P_j = \prod_{g \in \mathcal{G}} (P_j^g)^{\alpha_g}.$$

B.3 Equilibrium Derivation (Sector Level)

B.3.1 Price

Supplier-side price offer (sector g).

$$\begin{aligned} G_{nij}^{g,m}(p) &\equiv \Pr(p_{nij}^{g,m}(\omega) \leq p) = \Pr\left(\frac{c_n^g}{z_{ni}^m(\omega)} \tau_{nj}^{g,m} \leq p\right) = \Pr\left(z_{ni}^m(\omega) \geq \frac{c_n^g \tau_{nj}^{g,m}}{p}\right) \\ &= 1 - \Pr\left(z_{ni}^m(\omega) < \frac{c_n^g \tau_{nj}^{g,m}}{p}\right) = 1 - \exp\left\{-T_{ni} \left(\frac{c_n^g \tau_{nj}^{g,m}}{p}\right)^{-\theta}\right\} \\ &= 1 - \exp\left\{-T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta} p^\theta\right\}. \end{aligned}$$

Cheapest price distribution that is accepted by the buyer (sector g).

$$\begin{aligned}
G_j^g(p) &= \Pr(p_j^g(\omega) \leq p) \\
&= \Pr\left(\min_{n,i,m}\{p_{nij}^{g,m}(\omega)\} \leq p\right) \\
&= 1 - \Pr\left(\min_{n,i,m}\{p_{nij}^{g,m}(\omega)\} \geq p\right) \\
&= 1 - \Pr\{p_{n1j}^{g,W}(\omega) \geq p, p_{n1j}^{g,B}(\omega) \geq p, \dots, p_{NS_Nj}^{g,W}(\omega) \geq p, p_{NS_Nj}^{g,B}(\omega) \geq p\} \\
&= 1 - \prod_{n \in N} \prod_{i \in S_n} \prod_{m \in \{W,B\}} (1 - G_{nij}^{g,m}(p)) \\
&= 1 - \exp\left\{-\Phi_j^g p^\theta\right\} \quad \text{where } \Phi_j^g \equiv \sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta}.
\end{aligned}$$

B.3.2 Sourcing probability (sector g)

$$\begin{aligned}
\Pi_{nij}^{g,W} &\equiv \Pr\left(p_{nij}^{g,W}(\omega) \leq \min_{(n'i',m') \neq (ni,W)} p_{n'i'j}^{g,m'}(\omega)\right) \\
&= \int_0^\infty \Pr\left(\min_{(n'i',m') \neq (ni,W)} p_{n'i'j}^{g,m'}(\omega) \geq p \mid p_{nij}^{g,W}(\omega) = p\right) dG_{nij}^{g,W}(p) \\
&= \int_0^\infty \prod_{(n'i',m') \neq (ni,W)} (1 - G_{n'i'j}^{g,m'}(p)) dG_{nij}^{g,W}(p) \\
&= \int_0^\infty \exp\left(-p^\theta \sum_{(n'i',m') \neq (ni,W)} \kappa_{n'i'j}^{g,m'}\right) \left[\theta p^{\theta-1} \kappa_{nij}^{g,W} \exp(-p^\theta \kappa_{nij}^{g,W})\right] dp \\
&= \int_0^\infty \theta p^{\theta-1} \kappa_{nij}^{g,W} \exp\left[-p^\theta \left(\sum_{(n'i',m') \neq (ni,W)} \kappa_{n'i'j}^{g,m'} + \kappa_{nij}^{g,W}\right)\right] dp \\
&= \kappa_{nij}^{g,W} \int_0^\infty \theta p^{\theta-1} \exp(-p^\theta \Phi_j^g) dp \\
&= \kappa_{nij}^{g,W} \left[-\frac{1}{\Phi_j^g} \exp(-p^\theta \Phi_j^g)\right]_0^\infty \\
&= \frac{\kappa_{nij}^{g,W}}{\Phi_j^g} \quad \text{where } \Phi_j^g \equiv \sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} \kappa_{nij}^{g,m} = \sum_{n \in N} \sum_{i \in S_n} \sum_{m \in \{W,B\}} T_{ni} (c_n^g \tau_{nj}^{g,m})^{-\theta}. \quad (\text{B.5})
\end{aligned}$$

B.3.3 Comparative Statics

Substituting (4.6) into the sourcing probabilities (4.3) and (4.4), the ratio of within- to between-firm exports becomes

$$X_{nij}^m = \alpha_g E_j \frac{\kappa_{nij}^m}{\Phi_{jg}}, \quad \kappa_{nij}^m = T_{ni} (c_{ng} \tau_{nij}^m)^{-\theta},$$

with

$$\tau_{nij}^m = o_{ng} \iota_{jg} d_{nj}^{\delta_m} \exp\{z'_{nj} \Gamma_m\}, \quad m \in \{W, B\},$$

where δ_m is common across sectors g , and $o_{ng}, \iota_{jg}, z_{nj}$ do not depend on d_{nj} .

Substituting τ_{nij}^m , we have

$$\kappa_{nij}^m = T_{ni} (c_{ng} o_{ng} \iota_{jg})^{-\theta} d_{nj}^{-\theta \delta_m} \exp\{-\theta z'_{nj} \Gamma_m\}.$$

Since $\Phi_{jg} = \sum_{n', i', m'} \kappa_{n' i' jg}^{m'}$ factorizes $\iota_{jg}^{-\theta}$, write $\Phi_{jg} = \iota_{jg}^{-\theta} \tilde{\Phi}_{jg}$ with

$$\tilde{\Phi}_{jg} = \sum_{n', i', m'} T_{n' i'} (c_{n' g} o_{n' g})^{-\theta} d_{n' j}^{-\theta \delta_{m'}} \exp\{-\theta z'_{n' j} \Gamma_{m'}\}.$$

Therefore

$$X_{nij}^m = \alpha_g E_j \frac{T_{ni} (c_{ng} o_{ng})^{-\theta}}{\tilde{\Phi}_{jg}} d_{nj}^{-\theta \delta_m} \exp\{-\theta z'_{nj} \Gamma_m\}.$$

Define

$$B_{jg} \equiv \alpha_g E_j \frac{T_{ni} (c_{ng} o_{ng})^{-\theta}}{\tilde{\Phi}_{jg}},$$

which is independent of the mode m . Then

$$X_{nij}^m = B_{jg} d_{nj}^{-\theta \delta_m} \exp\{-\theta z'_{nj} \Gamma_m\}.$$

Define the aggregate (non-sectoral) within/arm's-length ratio

$$R_{nij}^{\text{agg}} \equiv \frac{\sum_g X_{nij}^W}{\sum_g X_{nij}^B} = \frac{\sum_g B_{jg} d_{nj}^{-\theta \delta_W} \exp\{-\theta z'_{nj} \Gamma_W\}}{\sum_g B_{jg} d_{nj}^{-\theta \delta_B} \exp\{-\theta z'_{nj} \Gamma_B\}}.$$

Because B_{jg} does not depend on m , it cancels across the sums:

$$R_{nij}^{\text{agg}} = d_{nj}^{-\theta(\delta_W - \delta_B)} \exp\{-\theta z'_{nj} (\Gamma_W - \Gamma_B)\}.$$

Taking logs and differentiating w.r.t. $\ln d_{nj}$:

$$\frac{\partial \ln R_{nij}^{\text{agg}}}{\partial \ln d_{nj}} = -\theta(\delta_W - \delta_B).$$

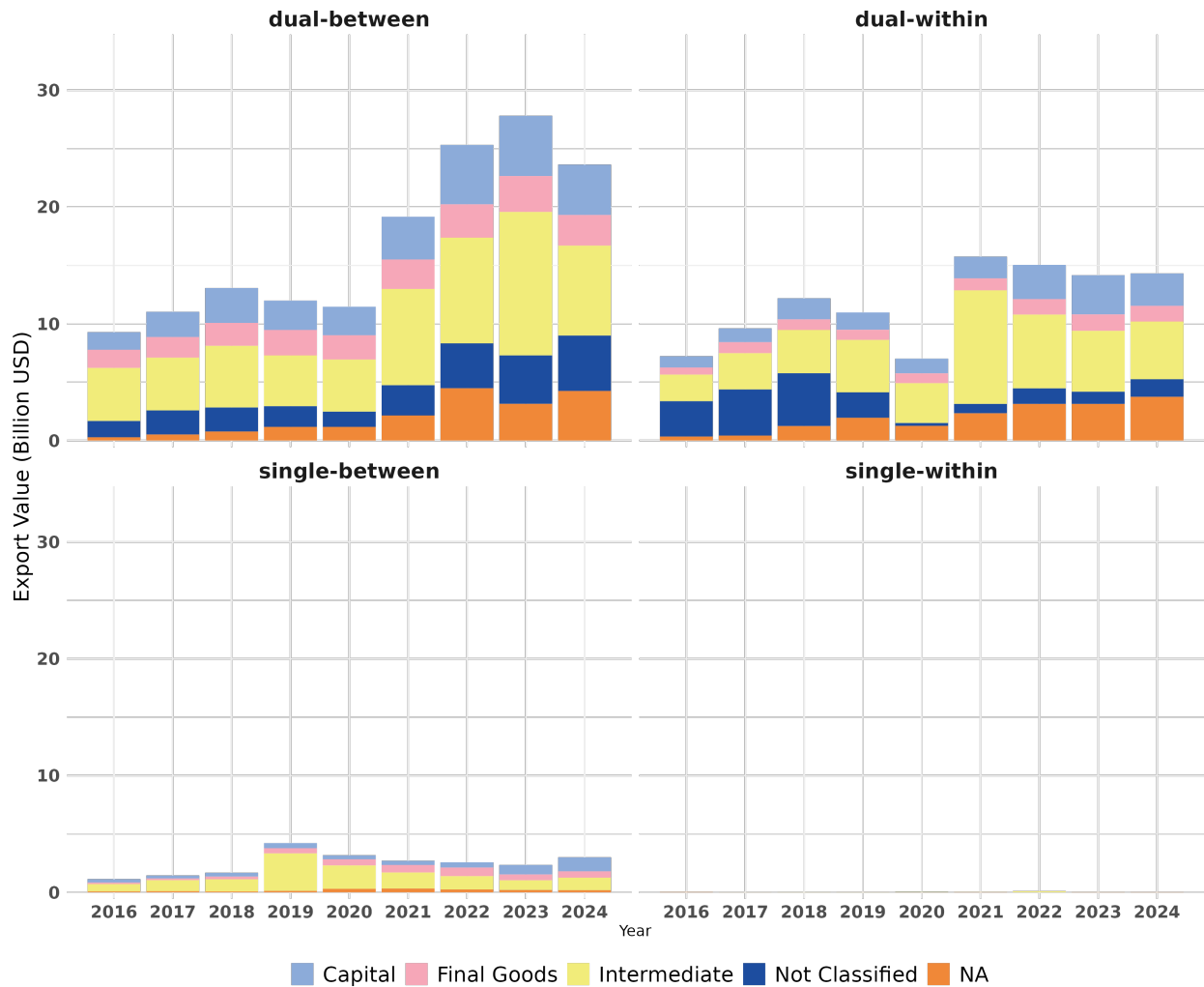
C Tables

Table A2: Countries by Distance Bin (India)

Distance Bin	Countries
1	United Arab Emirates, Nepal, Sri Lanka, Bhutan, Kuwait, Bangladesh, Bahrain, Qatar, Pakistan, Azerbaijan, Oman, Iran, Maldives, Myanmar, Turkmenistan, Uzbekistan, Laos, Tajikistan, Kazakhstan, Kyrgyzstan
2	Djibouti, Thailand, Israel, Hong Kong, Saudi Arabia, Singapore, Malaysia, Yemen, Vietnam, Iraq, Jordan, Cambodia, Lebanon, Mongolia, Georgia, Cyprus, Armenia, Macau, Eritrea, Syria
3	Indonesia, Somalia, Romania, Russia, Bulgaria, Ukraine, Egypt, Turkey, China, Lithuania, South Korea, Greece, Taiwan, Belarus, Philippines, Ethiopia, Seychelles, Latvia, Sudan, North Korea
4	Mauritius, Kenya, Tanzania, Germany, Slovenia, Serbia, Albania, Poland, Uganda, Croatia, Austria, Czech Republic, Sweden, Finland, Slovakia, Comoros, Estonia, Hungary, North Macedonia, Montenegro
5	Netherlands, Belgium, France, Japan, Italy, Denmark, Switzerland, Libya, Liechtenstein, Rwanda, Norway, Malta, Madagascar, Tunisia, Burundi, Réunion, Luxembourg, Monaco, San Marino
6	United Kingdom, Zambia, Portugal, Spain, Mozambique, Cameroon, Ireland, Zimbabwe, Chad, Algeria, Equatorial Guinea, Malawi, Gabon, Iceland, Central African Republic, Niger, Eswatini, Gibraltar, Andorra
7	South Africa, Nigeria, Mali, Sierra Leone, Guinea, Togo, Benin, Botswana, Burkina Faso, Ghana, Papua New Guinea, Angola, Lesotho, Morocco, Liberia, Namibia, Mauritania, Guinea-Bissau, Greenland
8	Canada, United States, New Zealand, Australia, Senegal, Fiji, Gambia, New Caledonia, Tonga, Bermuda, Vanuatu, Solomon Islands, Cape Verde, Anguilla, Montserrat, Marshall Islands, Kiribati, Tuvalu, Nauru
9	Suriname, Dominican Republic, Guyana, Jamaica, Venezuela, Cayman Islands, Haiti, Cuba, Bahamas, Saint Lucia, Netherlands Antilles, Guadeloupe, Barbados, Aruba, Grenada, Dominica, Martinique, Cook Islands, French Guiana
10	Falkland Islands, Mexico, Costa Rica, Brazil, Chile, Colombia, Ecuador, Guatemala, Panama, Paraguay, Peru, Argentina, Uruguay, Nicaragua, Honduras, El Salvador, Bolivia, Belize, French Polynesia

D Figures

Figure A7: Export Volume (Value) by Foreign MNE Affiliates



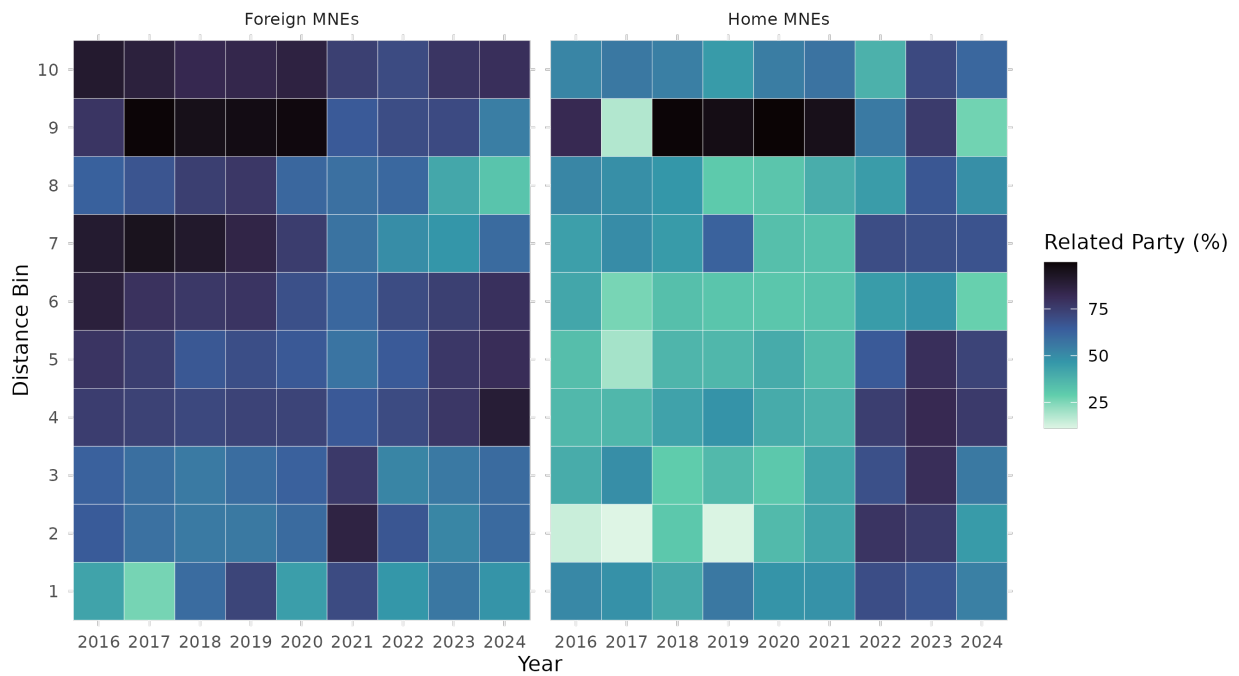
Notes: The y-axis is measured in billions of U.S. dollars. Classification is based on the concordance between HS codes and the Broad Economic Categories (BEC Rev. 4), which are then aggregated into three basic end-use categories defined by the United Nations System of National Accounts (SNA): capital goods, intermediate goods, and final (consumption) goods. Capital goods include machinery, equipment, dwellings, and other durable assets that are less disposable. “Not Classified” refers to items such as motor spirits (gasoline) and passenger motor vehicles, while “NA” indicates unmatched observations. In the labels, “within” denotes related-party export and “between” denotes arm’s-length export. The category labeled “dual” indicates firms engaged in both types of exports (as shown in the left panel (Foreign MNEs) in Figure 3), whereas “single” indicates firms operating in only one mode, either related-party or arm’s-length exports. The category labeled “single-within” contains non-zero values that are not visible in the figure due to the axis scale.

Figure A8: Export Value by Home MNE Affiliate Establishments



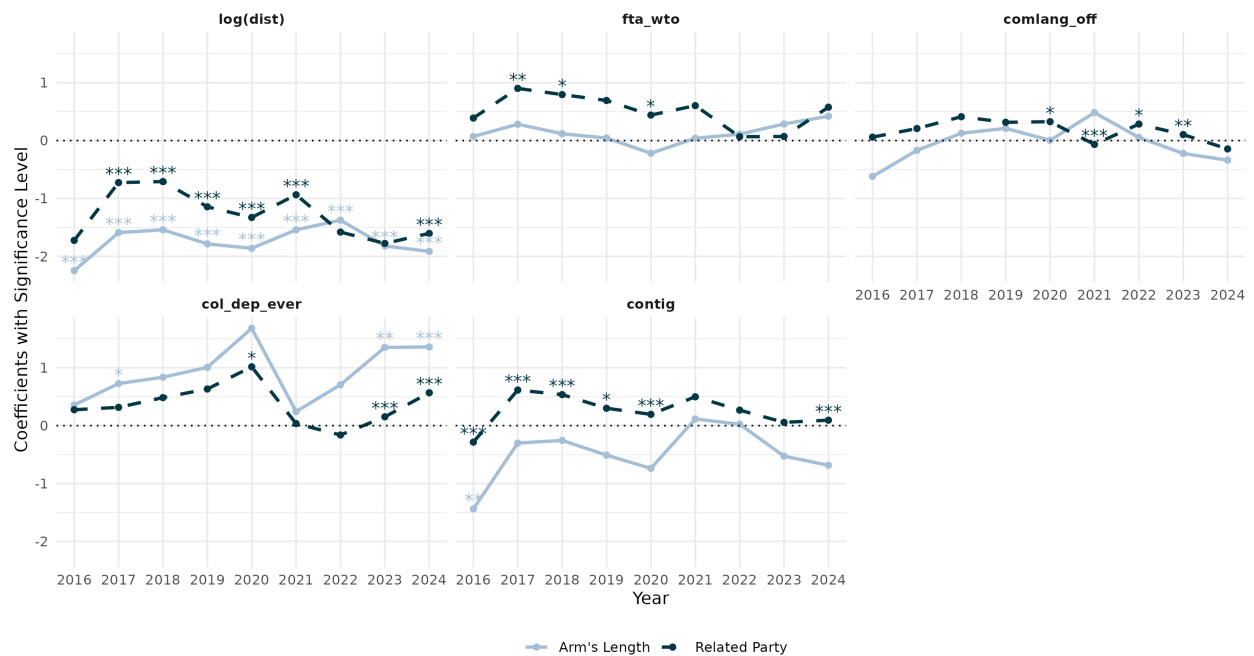
Notes: The y-axis is measured in billions of U.S. dollars. Classification is based on the concordance between HS codes and the Broad Economic Categories (BEC Rev. 4), which are then aggregated into three basic end-use categories defined by the United Nations System of National Accounts (SNA): capital goods, intermediate goods, and final (consumption) goods. Capital goods include machinery, equipment, dwellings, and other durable assets that are less disposable. “Not Classified” refers to items such as motor spirits (gasoline) and passenger motor vehicles, while “NA” indicates unmatched observations. In the labels, “within” denotes related-party trade and “between” denotes arm’s-length trade. The category labeled “dual” indicates firms engaged in both types of exports (as shown in the right panel (Home MNEs) in [Figure 3](#)), whereas “single” indicates firms operating in only one mode—either related-party or arm’s-length exports. The category labeled “single-within” contains non-zero values that are not visible in the figure due to the axis scale.

Figure A9: Related Party Trade Share by Distance Bin and Year - India, firm level weighted average



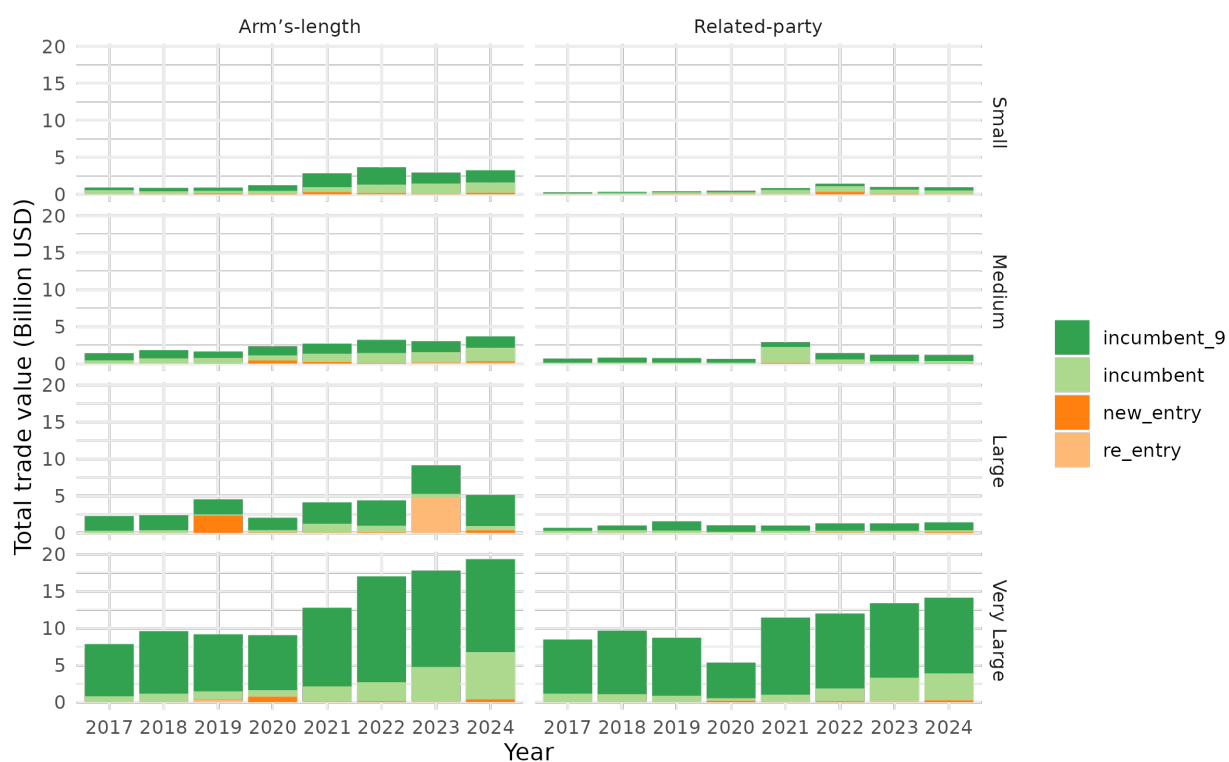
Notes: This figure illustrates the related-party trade share by distance for MNE firms headquartered outside of India versus those headquartered in India. The ratio is first calculated at the firm level. To obtain the average for each bin, I take a weighted mean where the weight is the firm's trade volume with each destination. In this way, destinations that account for more trade have a greater influence on the bin-level average.

Figure A10: Heterogeneous Effects of Gravity Variables on Related-Party vs. Arm's-Length Exports



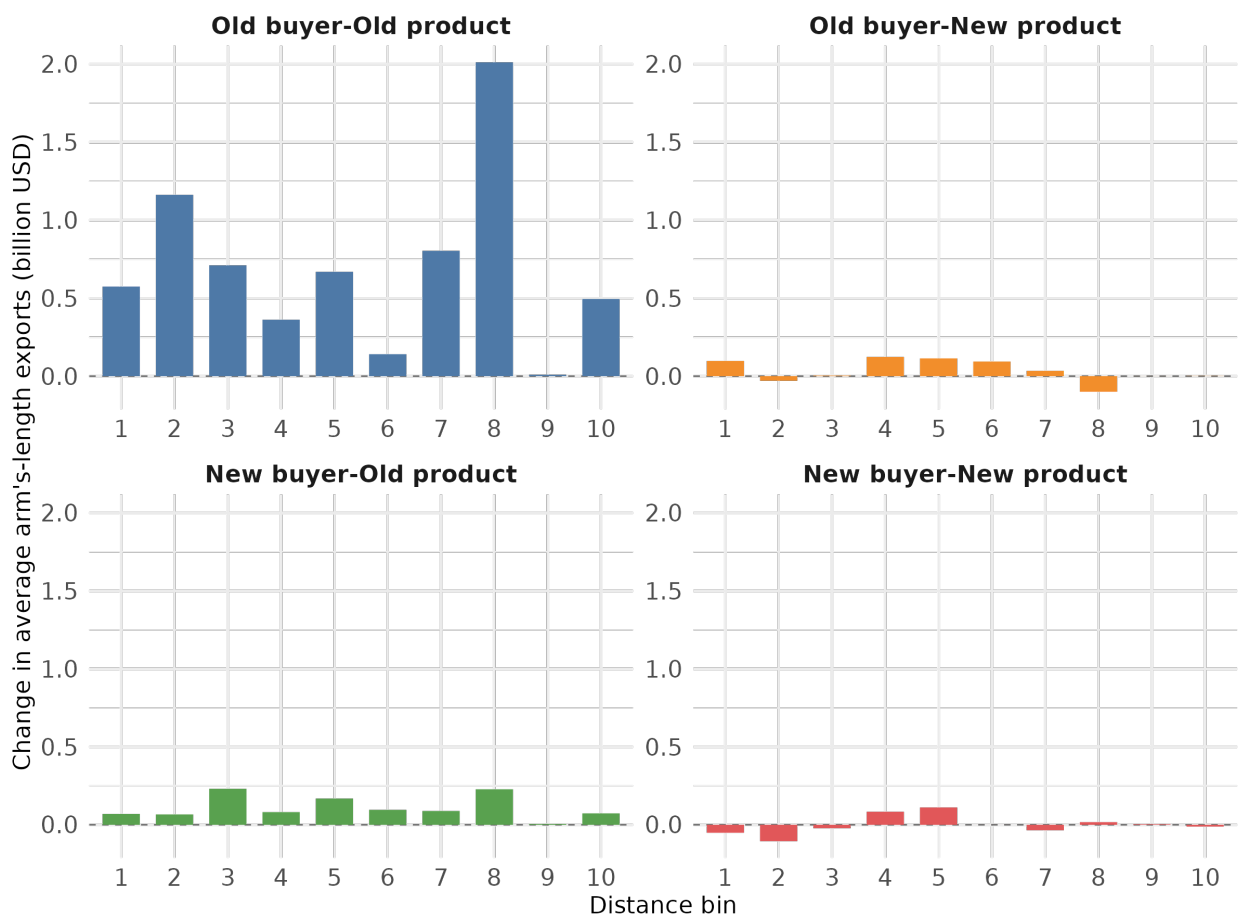
Notes: The regressors include an interaction term between gravity variables and a within-transfer dummy. The baseline category is arm's-length trade. P-value significance levels are indicated as follows: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. P-value labels on within-firm trade indicate the significance of the interaction term only, not the linear combination of the baseline and interaction. Robust standard errors are multi-way clustered by MNE affiliate, source country, product, and destination.

Figure A11: MNEs' Total Exports by Parent Firm Size and Entry Type: India



Notes: Entry type is classified into four categories: MNEs present throughout the sample period (incumbent_9), previously existing firms (incumbent), new MNEs (new_entry), and MNEs that had previously exited the export market and later re-entered (re_entry). Parent firm size is classified into four categories based on the number of countries in which the MNE has affiliates: small (1–5), medium (6–10), large (11–30), and very large (> 30).

Figure A12: Change in Average Arm’s-Length Exports, 2021–2024 Relative to 2017–2020 (Billion USD): India



Notes: The figure decomposes the change in average arm’s-length exports before and after 2021 across distance bins into four buyer-product relationship categories: old buyer-old product, old buyer-new product, new buyer-old product, and new buyer-new product. Each panel corresponds to one category, and within each panel the bars show the change in average export value for that category across distance bins. The dashed horizontal line marks zero, so bars above zero indicate an increase in average exports after 2021 relative to the pre-2021 period, while bars below zero indicate a decline.